
Restarted Reflected Halpern Acceleration for Augmented Primal–Dual Methods

Benqi Liu · Ju Cao · Wotao Yin ·
Zaiwen Wen

Abstract We study linearly constrained composite convex optimization with a smooth term having Lipschitz continuous gradient and a proximable non-smooth term. We develop a unified augmented primal–dual framework in which different metric choices recover primal–dual hybrid gradient and augmented Chambolle–Pock-type schemes, and also lead to a new fully augmented Chambolle–Pock family that retains the augmented quadratic term explicitly. For the exact scheme, we prove an exact degenerate preconditioned proximal-point reformulation; for the corresponding linearized scheme, we prove an exact preconditioned forward–backward splitting reformulation. These operator representations support reflected Halpern acceleration: the generated shadow iterates converge to Karush–Kuhn–Tucker points and satisfy nonergodic $O(1/k)$ bounds for the KKT residual and the objective gap. A scalar quadratic example shows that this global residual rate is worst-case tight. We further show that, for reflected Halpern trajectories, finite identification is a property of the shadow sequence rather than of the anchored Halpern state. Under an exact affine-face model after identification, we derive an exact reduced residual identity, a reduced Jacobian criterion for local sharpness, and a transfer from reduced error bounds to the original augmented KKT mapping. Under sharp-

Correspondence to: Zaiwen Wen.

Benqi Liu
Beijing International Center for Mathematical Research, Peking University, Beijing, China
E-mail: bqliu@pku.edu.cn

Ju Cao
School of Mathematical Sciences, Peking University, Beijing, China
E-mail: 2300010612@stu.pku.edu.cn

Wotao Yin
Alibaba US, DAMO Academy
E-mail: wotao.yin@alibaba-inc.com

Zaiwen Wen
Beijing International Center for Mathematical Research, Peking University, Beijing, China
E-mail: wenzw@pku.edu.cn

ness on the relevant set of restart points, we establish fixed-frequency and adaptive restart guarantees. When this sharpness is supplied by local metric subregularity or by a Hoffman–Robinson error bound, the resulting linear convergence is local: it applies to the tail of the trajectory after entry into the error-bound neighborhood. In the positive definite metric setting, this verification covers the identified regime and box linear and quadratic programs. Computational experiments on linear programming and convex quadratic programming benchmarks illustrate targeted regimes in which the proposed augmented variants can improve over established primal–dual baselines.

Keywords Augmented primal–dual methods · Halpern iteration · restart · active-set identification · sharpness

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1 Introduction

We study the linearly constrained composite convex optimization problem

$$\min_{x \in \mathbb{R}^n} \Phi(x) := f(x) + g(x) \quad \text{s.t.} \quad Ax = b, \quad (1)$$

where $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is convex and continuously differentiable with Lipschitz continuous gradient, and $g : \mathbb{R}^n \rightarrow (-\infty, +\infty]$ is proper, closed, convex, and proximal. We assume throughout that the feasible set is nonempty and that (1) has at least one optimal solution. This formulation includes, for example, equality-form linear programs, regularized least-squares models with linear side constraints, and box-constrained convex quadratic programs written with explicit equality constraints.

The paper is concerned with first-order primal–dual methods built from the augmented Lagrangian

$$L_\sigma(x, y) = f(x) + g(x) - \langle y, Ax - b \rangle + \frac{\sigma}{2} \|Ax - b\|^2, \quad \sigma \geq 0.$$

Augmentation is attractive both analytically and computationally: it injects curvature in the constraint directions and can enlarge the admissible primal–dual parameter region. At the same time, it complicates the usual complexity analysis. In particular, for equality-constrained augmented Lagrangian formulations the dual domain is unbounded; consequently, saddle-point gap estimates are not a satisfactory substitute for convergence guarantees stated directly in KKT residuals and objective-value gaps. The central question of this paper is therefore how to obtain global nonergodic KKT rates and local restart guarantees for a broad augmented primal–dual family.

Two lines of work motivate our approach. The first is the augmented-Lagrangian and proximal-point tradition. Rockafellar interpreted augmented Lagrangian methods as proximal point algorithms [32, 33]; Eckstein and Bertsekas then connected ADMM with Douglas–Rachford splitting [14]. See also

the survey [5]. Semi-proximal and generalized ADMM-type methods have subsequently become standard tools for structured convex optimization and conic programming [34, 16, 36]. The second line is matrix-free primal–dual splitting. It includes the Chambolle–Pock/primal–dual hybrid gradient (PDHG) method and related schemes for composite saddle-point models [7, 8, 15, 30]. Further developments treat more general primal–dual structures [12, 11]. These methods have cheap iterations and are well suited for large-scale implementations, but their classical convergence theory is usually formulated for saddle-point gaps or averaged quantities.

Recent developments have begun to bridge these viewpoints. Zhu et al. [37] proposed a unified augmented primal–dual framework that includes updates of the PDHG, Chambolle–Pock, and OGD types and established ergodic $O(1/N)$ guarantees without assuming bounded optimal multipliers. Degenerate preconditioned proximal point algorithms were studied systematically in [6]. Building on this operator-theoretic perspective, Sun et al. [35] reformulated a class of preconditioned ADMM schemes [36] as degenerate proximal point mappings and applied Halpern-type acceleration [18, 23] to obtain non-ergodic $O(1/k)$ KKT residual bounds. The related solver HPR-LP [9] shows that this viewpoint can also be computationally useful. These results show the power of the operator viewpoint, but they do not yet provide a single treatment of exact and linearized augmented primal–dual schemes, nor do they explain how reflected Halpern acceleration should be restarted in the local regime.

The local high-accuracy regime is central for first-order methods. Modern large-scale solvers for LP and QP, including PDLP and its extensions [1, 2], cuPDLP.jl and cuPDLPx [26, 24], HPR-LP [9], PDQP [28], HPR-QP [10], and PDHCG-II [22], rely on restart, scaling, and local residual reduction to reach useful accuracy. Theoretical progress in this direction is strongest for LP: sharpness-based restart analyses and refined PDHG geometry were developed in [3, 25, 27]. Local identification and error-bound theory provide a natural language for extending such ideas. Classical results of Hoffman and Robinson [20, 31], together with the active-manifold and partial-smoothness literature [21, 19], describe how residuals behave after the active structure has stabilized. Recent work of Díaz et al. [13] further clarifies active-set identification and rapid local convergence for degenerate primal–dual trajectories. What is still missing is a framework connecting these local geometric ideas with augmented primal–dual reflected Halpern schemes for the composite model (1).

This paper develops such a framework. The analysis moves from augmented primal–dual maps to dPPM and PFBS representations, reflected Halpern KKT rates, shadow identification, and restart under sharpness.

We introduce a unified augmented primal–dual scheme. Its metric choices recover updates of the PDHG type and the augmented Chambolle–Pock family (CP-AL), and also yield a new fully augmented Chambolle–Pock family (FA-CP), in which the augmented quadratic term is retained explicitly. For the exact scheme, we prove an exact degenerate preconditioned proximal-point reformulation. For the corresponding linearized scheme, we prove an exact preconditioned forward–backward splitting reformulation. These two operator

representations allow a common reflected Halpern analysis, yield nonergodic $O(1/k)$ bounds in KKT residuals and objective gaps, and provide the fixed-point objects needed for finite identification and restart.

We use several existing tools, including abstract Halpern residual estimates, restart under sharpness, partial smoothness, and Hoffman–Robinson error bounds. The contributions of the paper lie in the augmented primal–dual operator constructions, the translation of the fixed-point estimates into KKT and objective estimates for (1), the shadow-sequence identification mechanism for reflected Halpern trajectories, and the transfer of local error-bound information to restart guarantees for the resulting primal–dual maps.

- We develop a unified augmented primal–dual family whose metric choices recover existing schemes of the PDHG and CP-AL types and yield the new FA-CP family. For this family, we prove exact operator representations: the exact scheme is a degenerate preconditioned proximal point method, while the linearized scheme is a preconditioned forward–backward splitting method.
- We derive verifiable metric admissibility conditions and an explicit admissible reflected-relaxation range for the linearized map. Combining these representations with reflected Halpern estimates gives convergence to KKT points and nonergodic $O(1/k)$ bounds for KKT residuals and objective gaps. A scalar quadratic example shows that the global residual rate is worst-case tight.
- We identify the correct finite-identification object for reflected Halpern trajectories: it is the shadow sequence, not the anchored Halpern state. Under an additional exact affine-face hypothesis, we derive an exact reduced residual identity, a reduced Jacobian criterion for local sharpness, and a transfer from reduced error bounds to the original augmented KKT mapping.
- We prove fixed-frequency and adaptive restart results for reflected fixed-point maps under sharpness. In the positive definite metric setting, the required sharpness condition is verified through metric subregularity, the identified reduced geometry, and a Hoffman–Robinson error-bound argument for box-constrained linear and convex quadratic programs. We also report GPU experiments illustrating the regimes predicted by the theory.

The assumptions are layered according to the role of each result. The exact dPPM representation in Section 2 allows a possibly degenerate preconditioner, with resolvent regularity imposed separately. The linearized PFBS analysis in Section 3 uses a positive definite metric and the condition $\mu_x > L_h/2$ to obtain averagedness and an admissible reflection range. The identification results in Section 4 are local around a limiting KKT point and use partial smoothness; the exact reduced residual identity further assumes an affine-face model. Finally, the restart results in Section 5 are conditional on sharpness on the set visited by the restart trajectory. They are global only when such sharpness holds globally; in the applications derived from metric subregularity or Hoffman–Robinson error bounds, they should be read as local or tail linear convergence statements.

The rest of the paper is organized as follows. Section 2 studies the exact augmented primal–dual scheme from the proximal-point viewpoint. Section 3 develops the forward–backward view of the linearized scheme. Section 4 studies shadow identification and reduced residuals. Section 5 establishes the restart theory and local linear convergence under sharpness. Section 6 presents the numerical experiments.

2 A Proximal-Point View of Augmented Primal–Dual Methods

In this section, we focus on the exact augmented Lagrangian primal–dual scheme. At this stage the two subproblems are understood through their first-order optimality conditions; the regularity assumptions that make the induced resolvent single-valued and globally Lipschitz continuous are imposed in Section 2.2. Motivated by the reflected primal–dual hybrid gradient update, in which the multiplier is evaluated at the extrapolated primal point $2x^{k+1} - x^k$, we study the following augmented primal–dual scheme:

$$\begin{aligned} x^{k+1} &= \arg \min_x \left\{ L_\sigma(x, y^k) + \frac{1}{2} \|x - x^k\|_P^2 \right\}, \\ y^{k+1} &= \arg \max_y \left\{ L_\sigma(2x^{k+1} - x^k, y) - \frac{1}{2} \|y - y^k\|_Q^2 \right\}. \end{aligned} \quad (2)$$

Here the primal step is performed on the augmented Lagrangian, while the dual step retains the reflected primal argument. Different choices of the metric operators P and Q then lead to different augmented primal–dual realizations, including PDHG-type and augmented Chambolle–Pock-type schemes. The key observation is that, despite their different appearances, these methods are generated by a single degenerate preconditioned proximal-point map. This exact operator representation is the mechanism through which we apply reflected Halpern acceleration and convert abstract fixed-point residual estimates into nonergodic $O(1/k)$ bounds for the KKT residual and objective-value gap of problem (1).

We therefore begin by recalling the degenerate preconditioned proximal-point framework and then identify (2) as an instance of it.

2.1 A Preconditioned Proximal Point Reformulation

Let $\mathcal{H}$ be a real Hilbert space, and consider the monotone inclusion problem

$$\text{find } w \in \mathcal{H} \quad \text{such that} \quad 0 \in \mathcal{T}w. \quad (3)$$

Given a bounded, self-adjoint, and positive semidefinite linear operator $\mathcal{M} : \mathcal{H} \rightarrow \mathcal{H}$, the associated preconditioned proximal point iteration for solving (3) is

$$w^{k+1} = (\mathcal{M} + \mathcal{T})^{-1} \mathcal{M} w^k. \quad (4)$$

When $\mathcal{M}$ is only positive semidefinite, (4) is often called a degenerate preconditioned proximal point method (dPPM). In this subsection, we use only the structural assumptions needed for the algebraic reformulation: P and Q are assumed to be self-adjoint and positive semidefinite. Stronger conditions ensuring admissibility and regularity of the associated resolvent will be introduced in the next subsection.

We next show that the augmented Lagrangian primal-dual scheme (2) admits an exact proximal-point-type reformulation. To this end, let $w = (x, y)$ and define

$$\mathcal{M} = \begin{pmatrix} P & A^\top \\ A & Q \end{pmatrix}, \quad (5)$$

and

$$\mathcal{T}(x, y) = \begin{pmatrix} \partial\Phi(x) - A^\top y + \sigma A^\top (Ax - b) \\ Ax - b \end{pmatrix}. \quad (6)$$

Proposition 1 *With $\mathcal{M}$ and $\mathcal{T}$ defined in (5)–(6), the optimality conditions of (2) are equivalent to*

$$\mathcal{M}w^k \in \mathcal{M}w^{k+1} + \mathcal{T}(w^{k+1}), \quad w^{k+1} = (x^{k+1}, y^{k+1}).$$

Consequently, whenever the corresponding resolvent is well defined and single-valued at $\mathcal{M}w^k$, the update is exactly given by (4). If, in addition, $\mathcal{M} \succeq 0$, this is a degenerate preconditioned proximal point representation.

Proof. The displayed inclusion is, by (5)–(6), equivalent to

$$\begin{cases} Px^k + A^\top y^k \in Px^{k+1} + \partial\Phi(x^{k+1}) + \sigma A^\top (Ax^{k+1} - b), \\ Ax^k + Qy^k = 2Ax^{k+1} + Qy^{k+1} - b. \end{cases}$$

Equivalently,

$$\begin{cases} 0 \in \partial\Phi(x^{k+1}) - A^\top y^k + \sigma A^\top (Ax^{k+1} - b) + P(x^{k+1} - x^k), \\ 0 = -A(2x^{k+1} - x^k) + b - Q(y^{k+1} - y^k). \end{cases}$$

The first relation is precisely the optimality condition of the x -subproblem in (2), while the second relation is exactly the optimality condition of the y -subproblem. Therefore (2) and the corresponding instance of (4) are equivalent. $\square$

Proposition 1 identifies (2) with the resolvent form underlying dPPM. Consequently, different metric choices produce exact specializations of the same operator-theoretic template. We use the following naming convention. With $\sigma = 0$ and diagonal Euclidean metrics, (2) reduces to the primal-dual hybrid gradient method (PDHG) for the equality-constrained saddle formulation. With $\sigma \geq 0$ and the shifted primal metric $P = \tau^{-1}I_n - \sigma A^\top A$, the augmented quadratic term is linearized in the primal subproblem; this gives the augmented Chambolle–Pock scheme (CP-AL) studied in [37]. Finally, if the primal metric

Table 1: Metric choices for the exact augmented primal–dual scheme.

Scheme	σ	P	Q
Primal–dual hybrid gradient (PDHG) [8]	$\sigma = 0$	$\tau^{-1}I_n$	$\rho^{-1}I_m$
Augmented Chambolle–Pock (CP-AL) [37]	$\sigma \geq 0$	$\tau^{-1}I_n - \sigma A^\top A$	$\rho^{-1}I_m$
Fully augmented Chambolle–Pock (FA-CP)	$\sigma \geq 0$	$\tau^{-1}I_n$	$\rho^{-1}I_m$

remains $\tau^{-1}I_n$, then the augmented quadratic term is retained explicitly in the primal minimization. We refer to this new specialization as the fully augmented Chambolle–Pock scheme (FA-CP). Table 1 summarizes the corresponding metric choices. The $\sigma = 0$ member of FA-CP coincides with the PDHG-type corner, while its positive- σ members are fully augmented. Thus FA-CP differs from CP-AL in the primal metric: CP-AL cancels the added quadratic through a shifted metric, whereas FA-CP keeps the augmented quadratic in the primal minimization. To the best of our knowledge, the positive- σ members of this fully augmented CP-type family have not been isolated and analyzed before for the composite model (1). In all three cases the dual update has the common extrapolated form $y^{k+1} = y^k - \rho(A(2x^{k+1} - x^k) - b)$.

2.2 Admissibility and Resolvent Regularity

Section 2.1 identifies (2) with a proximal-point-type resolvent representation. We now impose verifiable conditions under which the associated resolvent is well defined and globally Lipschitz continuous. These conditions also guarantee that the metric operator $\mathcal{M}$ is an admissible preconditioner for $\mathcal{T}$ in the sense recalled below.

We first recall the notion of an admissible preconditioner.

Definition 1 (Admissible preconditioner) Let $\mathcal{T} : \mathcal{H} \rightrightarrows \mathcal{H}$ be a set-valued operator. A linear, bounded, self-adjoint, and positive semidefinite operator $\mathcal{M} : \mathcal{H} \rightarrow \mathcal{H}$ is called an *admissible preconditioner* for $\mathcal{T}$ if the mapping $\widehat{\mathcal{T}} := (\mathcal{M} + \mathcal{T})^{-1}\mathcal{M}$ is single-valued and has full domain.

We now specialize to the operator pair associated with the augmented Lagrangian primal–dual scheme.

Assumption 1. Let $(\mathcal{M}, \mathcal{T})$ be the operator pair defined in (5)–(6). Assume that

- (i) Q is self-adjoint and positive definite;
- (ii) $\mathcal{M} \succeq 0$;
- (iii) the operator $G := \partial\Phi + P + \sigma A^\top A$ has a single-valued inverse $G^{-1} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ that is globally Lipschitz continuous.

Assumption 1 is readily verified from the primal metric. Condition (iii) is the additional primal regularity used below to solve the resolvent. A convenient sufficient condition is

$$P + \sigma A^\top A \succeq \mu I \quad \text{for some } \mu > 0.$$

Indeed, then $G = \partial\psi$, where

$$\psi(x) := \Phi(x) + \frac{1}{2} \langle x, (P + \sigma A^\top A)x \rangle.$$

The function ψ is μ -strongly convex. Therefore, for every $\xi \in \mathbb{R}^n$, the function $x \mapsto \psi(x) - \langle \xi, x \rangle$ has a unique minimizer; equivalently, $\xi \in Gx$ for a unique x and G^{-1} has full domain. If $p_i \in Gx_i$ for $i = 1, 2$, strong monotonicity gives

$$\mu \|x_1 - x_2\|^2 \leq \langle p_1 - p_2, x_1 - x_2 \rangle \leq \|p_1 - p_2\| \|x_1 - x_2\|,$$

and therefore $\|G^{-1}p_1 - G^{-1}p_2\| \leq \mu^{-1} \|p_1 - p_2\|$. Thus Assumption 1 holds whenever the primal metric plus the augmented quadratic term is uniformly positive definite; for the metric choices in Table 1, this reduces to the familiar step-size restrictions discussed below.

The next result records the corresponding resolvent regularity and admissibility.

Proposition 2 *Under Assumption 1, the resolvent $(\mathcal{M} + \mathcal{T})^{-1}$ is single-valued, everywhere defined, and globally Lipschitz continuous on $\mathbb{R}^n \times \mathbb{R}^m$. Moreover, the operator $\widehat{\mathcal{T}} := (\mathcal{M} + \mathcal{T})^{-1}\mathcal{M}$ is single-valued and has full domain, and $\mathcal{M}$ is an admissible preconditioner for $\mathcal{T}$.*

Proof. Let $v = (\xi, \zeta) \in \mathbb{R}^n \times \mathbb{R}^m$, and suppose that $w = (x, y)$ satisfies $v \in (\mathcal{M} + \mathcal{T})(x, y)$. By the definitions of $\mathcal{M}$ and $\mathcal{T}$, this is equivalent to

$$\begin{cases} \xi \in Px + \partial\Phi(x) + \sigma A^\top (Ax - b), \\ \zeta = 2Ax + Qy - b. \end{cases}$$

It follows that

$$x = G^{-1}(\xi + \sigma A^\top b), \tag{7}$$

which is uniquely determined by Assumption 1(iii). Moreover,

$$y = Q^{-1}(\zeta + b - 2Ax), \tag{8}$$

which is also uniquely determined because Q is positive definite. Conversely, for every $v = (\xi, \zeta)$, the pair (x, y) defined by (7)–(8) satisfies the displayed system above. Therefore, $(\mathcal{M} + \mathcal{T})^{-1}$ is single-valued and everywhere defined.

Let L_G be a Lipschitz constant of G^{-1} . To prove Lipschitz continuity, let $v_i = (\xi_i, \zeta_i)$ and $w_i = (x_i, y_i) := (\mathcal{M} + \mathcal{T})^{-1}v_i$ for $i = 1, 2$. From (7) and the Lipschitz continuity of G^{-1} ,

$$\|x_1 - x_2\| \leq L_G \|\xi_1 - \xi_2\|.$$

Since $\|\xi_1 - \xi_2\| \leq \|v_1 - v_2\|$, this yields

$$\|x_1 - x_2\| \leq L_G \|v_1 - v_2\|.$$

Using (8), we obtain $y_1 - y_2 = Q^{-1}((\zeta_1 - \zeta_2) - 2A(x_1 - x_2))$, and thus

$$\|y_1 - y_2\| \leq \|Q^{-1}\|(\|\zeta_1 - \zeta_2\| + 2\|A\|\|x_1 - x_2\|) \leq \|Q^{-1}\|(1 + 2\|A\|L_G)\|v_1 - v_2\|.$$

Therefore,

$$\|w_1 - w_2\|^2 = \|x_1 - x_2\|^2 + \|y_1 - y_2\|^2 \leq L_{\text{res}}^2 \|v_1 - v_2\|^2,$$

where

$$L_{\text{res}} := \left[L_G^2 + \|Q^{-1}\|^2 (1 + 2\|A\|L_G)^2 \right]^{1/2}.$$

This proves that $(\mathcal{M} + \mathcal{T})^{-1}$ is globally Lipschitz continuous.

Since $\mathcal{M}$ is a bounded linear operator, the composition $\widehat{\mathcal{T}} = (\mathcal{M} + \mathcal{T})^{-1}\mathcal{M}$ is single-valued and defined on the whole space. Since, by assumption, $\mathcal{M}$ is self-adjoint and positive semidefinite, the conclusion follows from Definition 1. $\square$

In our finite-dimensional setting, the remaining structural assumptions needed for the abstract reflected Halpern theory are also automatic.

Proposition 3 *Under Assumption 1, the operator $\mathcal{T}$ is maximal monotone and $\text{ran}(\mathcal{M})$ is closed.*

Proof. Define $\widetilde{\Phi} : \mathbb{R}^n \times \mathbb{R}^m \rightarrow (-\infty, +\infty]$ by $\widetilde{\Phi}(x, y) := \Phi(x)$. Then

$$\partial \widetilde{\Phi}(x, y) = \begin{pmatrix} \partial \Phi(x) \\ 0 \end{pmatrix},$$

Therefore $\partial \widetilde{\Phi}$ is maximal monotone. Next define the linear operator

$$\mathcal{K}(x, y) := \begin{pmatrix} \sigma A^\top Ax - A^\top y \\ Ax \end{pmatrix}.$$

For any $w_i = (x_i, y_i)$, $i = 1, 2$, one has

$$\langle \mathcal{K}w_1 - \mathcal{K}w_2, w_1 - w_2 \rangle = \sigma \|A(x_1 - x_2)\|^2 \geq 0,$$

Thus $\mathcal{K}$ is monotone. Since $\mathcal{K}$ is linear, bounded, and everywhere defined, the sum $\partial \widetilde{\Phi} + \mathcal{K}$ is maximal monotone. Finally,

$$\mathcal{T} = \partial \widetilde{\Phi} + \mathcal{K} - \begin{pmatrix} \sigma A^\top b \\ b \end{pmatrix},$$

Thus $\mathcal{T}$ is maximal monotone as well. Because $\mathcal{M}$ is a linear operator on the finite-dimensional space $\mathbb{R}^n \times \mathbb{R}^m$, its range is a linear subspace and is therefore closed. $\square$

Remark 1 (Step-size conditions) For the representative schemes in Section 2.1, Assumption 1 reduces to familiar step-size conditions. For PDHG, $P = \tau^{-1}I_n$, $Q = \rho^{-1}I_m$, and $\sigma = 0$, with $P + \sigma A^\top A = \tau^{-1}I_n$; the condition $\mathcal{M} \succeq 0$ reduces, by the Schur complement, to $\tau\rho\|A\|^2 \leq 1$, which is precisely the classical PDHG step-size condition; see [8]. For CP-AL, $P = \tau^{-1}I_n - \sigma A^\top A$ and $Q = \rho^{-1}I_m$, and therefore $P + \sigma A^\top A = \tau^{-1}I_n$, while $\mathcal{M} \succeq 0$ reduces to $\tau(\sigma + \rho)\|A\|^2 \leq 1$, which is precisely the CP-AL step-size condition; see [37]. For FA-CP, $P = \tau^{-1}I_n$ and $Q = \rho^{-1}I_m$, which gives $P + \sigma A^\top A \succeq \tau^{-1}I_n$, and $\mathcal{M} \succeq 0$ again yields $\tau\rho\|A\|^2 \leq 1$. In all three cases, $P + \sigma A^\top A \succeq \tau^{-1}I_n$; consequently Assumption 1(iii) follows from the sufficient condition above. Thus the abstract assumptions recover the standard step-size restriction for PDHG, the natural one for CP-AL, and a PDHG-type condition for the fully augmented members of FA-CP.

2.3 Reflected Halpern Acceleration and $O(1/k)$ KKT Rates

Given an admissible preconditioner $\mathcal{M}$, set $\widehat{\mathcal{T}} := (\mathcal{M} + \mathcal{T})^{-1}\mathcal{M}$. We consider the reflected Halpern iteration

$$w^{k+1} = \frac{1}{k+2}w^0 + \frac{k+1}{k+2}\left((1+\gamma)\widehat{\mathcal{T}}(w^k) - \gamma w^k\right), \quad \gamma \in (-1, 1]. \quad (9)$$

Equivalently, defining

$$\hat{w}^k := \widehat{\mathcal{T}}(w^k), \quad \bar{w}^k := (1+\gamma)\hat{w}^k - \gamma w^k,$$

we may rewrite (9) as

$$w^{k+1} = \frac{1}{k+2}w^0 + \frac{k+1}{k+2}\bar{w}^k.$$

Following the reflected Halpern acceleration framework for dPPM mappings in [35], Algorithm 1 records the iteration in the shadow–reflection–anchoring form used throughout the paper. The shadow sequence $\{\hat{w}^k\}$, rather than only the anchored sequence $\{w^k\}$, is the object on which the KKT residual estimates are stated.

Algorithm 1: Reflected Halpern-Accelerated dPPM

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1: Input:  $w^0 \in \mathcal{H}$  and  $\gamma \in (-1, 1]$ .
2: for  $k = 0, 1, 2, \dots$  do
3:   Step 1.  $\hat{w}^k = \widehat{\mathcal{T}}(w^k)$ .
4:   Step 2.  $\bar{w}^k = (1+\gamma)\hat{w}^k - \gamma w^k$ .
5:   Step 3.  $w^{k+1} = (w^0 + (k+1)\bar{w}^k)/(k+2)$ .
6: end for
7: Output: shadow sequence  $\{\hat{w}^k\}$ .

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We now invoke the abstract reflected Halpern theory for dPPM mappings. The following proposition is a direct specialization of [35, Theorem 2.7, Proposition 2.9] and records the statements needed below.

Proposition 4 Let $\mathcal{T} : \mathcal{H} \rightrightarrows \mathcal{H}$ be a maximal monotone operator with $\mathcal{T}^{-1}(0) \neq \emptyset$, and let $\mathcal{M}$ be an admissible preconditioner for $\mathcal{T}$ with closed range. Let $\widehat{\mathcal{T}} := (\mathcal{M} + \mathcal{T})^{-1}\mathcal{M}$, and let the sequences $\{w^k\}$, $\{\hat{w}^k\}$, and $\{\bar{w}^k\}$ be generated by Algorithm 1. Then the following statements hold.

- (i) If $\gamma \in (-1, 1]$ and $(\mathcal{M} + \mathcal{T})^{-1}$ is continuous, then the sequence $\{\hat{w}^k\}$ converges strongly to a point $w^* \in \mathcal{T}^{-1}(0)$. Moreover, if $\gamma \in (-1, 1)$, then the sequences $\{w^k\}$ and $\{\bar{w}^k\}$ also converge strongly to the same limit w^* .
- (ii) If $\gamma \in (-1, 1]$, the relaxed residual satisfies

$$\|w^k - \bar{w}^k\|_{\mathcal{M}} \leq \frac{2\|w^0 - w^*\|_{\mathcal{M}}}{k+1}, \quad \forall k \geq 0, \forall w^* \in \mathcal{T}^{-1}(0).$$

Equivalently,

$$\|w^k - \hat{w}^k\|_{\mathcal{M}} \leq \frac{2}{(1+\gamma)(k+1)}\|w^0 - w^*\|_{\mathcal{M}}, \quad \forall k \geq 0, \forall w^* \in \mathcal{T}^{-1}(0).$$

Proof. Choose a decomposition $\mathcal{M} = CC^*$ as in [6, Proposition 2.3]. In the notation of [35], take $\alpha = 2$ and $\rho = 1 + \gamma$. Our $\hat{w}^k$ corresponds to their $\bar{w}^k$, while our $\bar{w}^k$ corresponds to their $\hat{w}^{k+1}$. Statement (i) follows from [35, Theorem 2.7(a)]. Statement (ii) follows from [35, Proposition 2.9], again with $\alpha = 2$ and $\rho = 1 + \gamma \in (0, 2]$. $\square$

Remark 2 (Euclidean residual in the Halpern case) The Lipschitz regularity in Proposition 2 also recovers the Euclidean residual estimate used in the spADMM analysis of [35]. Specifically, assume that $(\mathcal{M} + \mathcal{T})^{-1}$ is L -Lipschitz continuous and let $\mathcal{M} = CC^*$. In the Halpern case $\gamma = 0$, Corollary 2.11 of [35] gives, for every $w^* \in \mathcal{T}^{-1}(0)$ and every $k \geq 0$,

$$\|w^k - \hat{w}^k\| \leq \frac{1}{k+1}\|w^0 - w^*\| + \frac{(5k+1)L\|C\|}{(k+1)^2}\|w^0 - w^*\|_{\mathcal{M}}.$$

Thus the Lipschitz resolvent condition is not merely a well-posedness device: in the nonreflected Halpern case it converts the degenerate metric estimate into an ambient Euclidean residual bound. The subsequent analysis uses the $\mathcal{M}$ -residual estimate in Proposition 4(ii), which is available throughout the reflected range $\gamma \in (-1, 1]$.

Remark 3 The endpoint $\gamma = -1$ is degenerate in the exact dPPM setting as well, since then $\bar{w}^k = (1 + \gamma)\hat{w}^k - \gamma w^k = w^k$, and the Halpern update reduces to $w^{k+1} = \frac{1}{k+2}w^0 + \frac{k+1}{k+2}w^k = w^0$. Accordingly, the abstract convergence and residual statements above are stated only for $\gamma > -1$.

We now specialize the abstract residual estimate to the augmented KKT operator and then convert it back to the original KKT residual. For $w = (x, y)$, define the original KKT mapping and its residual by

$$\mathcal{R}(x, y) := \begin{pmatrix} \nabla f(x) + \partial g(x) - A^\top y \\ Ax - b \end{pmatrix}, \quad r_{\text{KKT}}(x, y) := \text{dist}(0, \mathcal{R}(x, y)). \quad (10)$$

Let $\mathcal{E}_\sigma$ denote the linear map $\mathcal{E}_\sigma(u, v) := (u - \sigma A^\top v, v)$. For a set S , write $\text{dist}_{\mathcal{M}}(w, S)$ for $\inf_{u \in S} \|w - u\|_{\mathcal{M}}$.

Theorem 1 *Suppose that Assumption 1 holds, that $\mathcal{T}^{-1}(0) \neq \emptyset$, and that $\gamma \in (-1, 1)$. Let $\{w^k\}$, $\{\hat{w}^k\}$, and $\{\bar{w}^k\}$ be generated by Algorithm 1. Then $\{w^k\}$ and $\{\hat{w}^k\}$ converge strongly to a common limit $w^* = (x^*, y^*) \in \mathcal{T}^{-1}(0) = \mathcal{R}^{-1}(0)$. Moreover, for every $k \geq 0$,*

$$r_{\text{KKT}}(\hat{w}^k) \leq \frac{2\|\mathcal{E}_\sigma\|\sqrt{\|\mathcal{M}\|}}{(1+\gamma)(k+1)} \text{dist}_{\mathcal{M}}(w^0, \mathcal{T}^{-1}(0)).$$

Consequently, $r_{\text{KKT}}(\hat{w}^k) = O(1/k)$. Set

$$C_* := \sup_{k \geq 0} (\|\hat{x}^k - x^*\| + \|\hat{y}^k\|) + \|y^*\|.$$

Then $C_* < \infty$ and

$$|\Phi(\hat{x}^k) - \Phi(x^*)| \leq C_* r_{\text{KKT}}(\hat{w}^k), \quad k \geq 0.$$

In particular, $|\Phi(\hat{x}^k) - \Phi(x^*)| = O(1/k)$.

Proof. By Propositions 3 and 2, all assumptions of Proposition 4 are satisfied. Since $\gamma \in (-1, 1)$, Proposition 4(i) implies that $w^k \rightarrow w^*$ and $\hat{w}^k \rightarrow w^*$ for some $w^* \in \mathcal{T}^{-1}(0)$. By the definition of $\mathcal{T}$, its second component is $Ax - b$. On the zero set of $\mathcal{T}$, the augmentation term therefore vanishes, and $\mathcal{T}^{-1}(0) = \mathcal{R}^{-1}(0)$.

For any (x, y) , the set-valued identity $\mathcal{R}(x, y) = \mathcal{E}_\sigma(\mathcal{T}(x, y))$ follows directly from the definition of $\mathcal{T}$: its first component differs from that of $\mathcal{R}$ by $\sigma A^\top(Ax - b)$, while the second components are identical. Therefore

$$r_{\text{KKT}}(\hat{w}^k) \leq \|\mathcal{E}_\sigma\| \text{dist}(0, \mathcal{T}(\hat{w}^k)).$$

Moreover, the resolvent relation for $\hat{w}^k$ gives $\mathcal{M}(w^k - \hat{w}^k) \in \mathcal{T}(\hat{w}^k)$, and therefore

$$\text{dist}(0, \mathcal{T}(\hat{w}^k)) \leq \|\mathcal{M}(w^k - \hat{w}^k)\| \leq \sqrt{\|\mathcal{M}\|} \|w^k - \hat{w}^k\|_{\mathcal{M}}.$$

Since $\bar{w}^k = (1 + \gamma)\hat{w}^k - \gamma w^k$, we have $w^k - \bar{w}^k = (1 + \gamma)(w^k - \hat{w}^k)$, and Proposition 4(ii), after taking the infimum over $\mathcal{T}^{-1}(0)$, yields

$$\|w^k - \hat{w}^k\|_{\mathcal{M}} = \frac{1}{1 + \gamma} \|w^k - \bar{w}^k\|_{\mathcal{M}} \leq \frac{2}{(1 + \gamma)(k + 1)} \text{dist}_{\mathcal{M}}(w^0, \mathcal{T}^{-1}(0)).$$

Combining the last three estimates proves the residual bound.

It remains to prove the objective estimate. The resolvent relation also implies $\mathcal{T}(\hat{w}^k) \neq \emptyset$, and therefore $\mathcal{R}(\hat{w}^k) \neq \emptyset$. Since $\mathcal{R}(\hat{w}^k)$ is closed and convex in finite dimensions, it contains a least-norm element. Let $r^k = (r_x^k, r_y^k) \in \mathcal{R}(\hat{w}^k)$ satisfy $\|r^k\| = r_{\text{KKT}}(\hat{w}^k)$. Then $r_y^k = A\hat{x}^k - b$ and $r_x^k \in \partial\Phi(\hat{x}^k) - A^\top \hat{y}^k$. Thus there exists $g^k \in \partial\Phi(\hat{x}^k)$ such that $g^k = A^\top \hat{y}^k + r_x^k$. By convexity of Φ ,

$$\Phi(\hat{x}^k) - \Phi(x^*) \leq \langle g^k, \hat{x}^k - x^* \rangle = \langle r_x^k, \hat{x}^k - x^* \rangle + \langle \hat{y}^k, A\hat{x}^k - b \rangle.$$

Consequently,

$$\Phi(\hat{x}^k) - \Phi(x^*) \leq (\|\hat{x}^k - x^*\| + \|\hat{y}^k\|) r_{\text{KKT}}(\hat{w}^k).$$

Since $w^* \in \mathcal{R}^{-1}(0)$, there exists $g^* \in \partial\Phi(x^*)$ such that $g^* = A^\top y^*$. Again by convexity,

$$\Phi(\hat{x}^k) - \Phi(x^*) \geq \langle g^*, \hat{x}^k - x^* \rangle = \langle y^*, A\hat{x}^k - b \rangle \geq -\|y^*\| r_{\text{KKT}}(\hat{w}^k).$$

Combining the upper and lower bounds gives

$$|\Phi(\hat{x}^k) - \Phi(x^*)| \leq (\|\hat{x}^k - x^*\| + \|\hat{y}^k\| + \|y^*\|) r_{\text{KKT}}(\hat{w}^k).$$

Since $\hat{w}^k \rightarrow w^*$, the constant C_* is finite. The stated $O(1/k)$ objective-gap bound follows from the residual estimate. $\square$

Remark 4 (Worst-case tightness) The preceding $O(1/k)$ residual estimate is sharp in the general convex setting. To see this, fix $\gamma \in (-1, 1)$ and choose $\eta \in (0, 1)$ with $(1 + \gamma)\eta \leq 1$. For the scalar problem

$$\min_{x \in \mathbb{R}} \frac{a}{2} x^2, \quad a := \frac{\eta}{1 - \eta},$$

viewed as the unconstrained instance of (1) and equipped with $\mathcal{M} = 1$, the dPPM map is $F(x) = (I + \mathcal{T})^{-1}x = (1 - \eta)x$ and $r_F(x) = |x - F(x)| = \eta|x|$. Set $q := 1 - (1 + \gamma)\eta$. Then the reflected map is $\mathcal{S}_\gamma(x) = qx$, and the scalar Halpern recursion becomes

$$x^{k+1} = \frac{1}{k+2}x^0 + \frac{k+1}{k+2}qx^k.$$

Multiplying by $k+2$ and setting $z_k := (k+1)x^k$ gives $z_{k+1} = x^0 + qz_k$ with $z_0 = x^0$. Therefore $z_k = (1 + q + \dots + q^k)x^0$. Starting from $x^0 \neq 0$, this yields

$$x^k = \frac{1 - q^{k+1}}{(1 + \gamma)\eta(k+1)}x^0, \quad r_F(x^k) = \frac{1 - q^{k+1}}{(1 + \gamma)(k+1)}|x^0|.$$

The shadow point satisfies $\hat{x}^k = F(x^k)$, and its KKT residual is $r_{\text{KKT}}(\hat{x}^k) = |a\hat{x}^k| = r_F(x^k)$. Thus

$$\lim_{k \rightarrow \infty} (k+1)r_{\text{KKT}}(\hat{x}^k) = \frac{|x^0|}{1 + \gamma} > 0,$$

This shows that the nonergodic residual bound cannot, in general, be improved to $o(1/k)$.

3 A Forward–Backward View of Linearized Reflected Schemes

This section studies linearized augmented primal–dual schemes. The linearized update admits an exact preconditioned forward–backward splitting (PFBS) representation, and its reflected relaxation range is controlled by the Schur complement quantity

$$\mu_x := \lambda_{\min}(P - A^\top Q^{-1}A).$$

This representation leads to reflected Halpern iterates with nonergodic $O(1/k)$ bounds for the KKT residual and the objective-value gap.

3.1 A Linearized Forward–Backward Reformulation

The computational motivation for the linearized schemes is to move a smooth component of the primal objective to the forward step, so that the shadow x -update becomes explicit or reduces to a simple proximal operation. This mechanism applies directly to the PDHG metric and to the shifted CP-AL metric. For instance, when $h = f$, the PDHG choice yields the usual forward–backward/proximal-gradient update, while the CP-AL metric cancels the augmented quadratic in the backward subproblem and leaves a proximal step for g . The same linearization does not play this role for the fully augmented FA-CP metric with $P = \tau^{-1}I$ and $\sigma > 0$: the term $\frac{\sigma}{2}\|Ax - b\|^2$ remains in the primal minimization, and the resulting subproblem is generally still coupled through A . Thus the linearized theory below is most useful computationally for the PDHG and CP-AL corners, while FA-CP is treated later as a subproblem-based augmented method.

To cover different linearization choices in a single notation, let $h : \mathbb{R}^n \rightarrow \mathbb{R}$ be convex and continuously differentiable, with L_h -Lipschitz continuous gradient, and assume that

$$\phi_h(x) := f(x) - h(x) + \frac{\sigma}{2}\|Ax - b\|^2 \quad \text{is convex on } \mathbb{R}^n. \quad (11)$$

Recall from (6) that

$$\mathcal{T}(x, y) = \begin{pmatrix} \partial g(x) + \nabla f(x) - A^\top y + \sigma A^\top (Ax - b) \\ Ax - b \end{pmatrix}.$$

We split $\mathcal{T}$ as

$$\begin{aligned} \mathcal{T} &= \mathcal{A}_h + \mathcal{B}_h, & \mathcal{B}_h(x, y) &:= \begin{pmatrix} \nabla h(x) \\ 0 \end{pmatrix}, \\ \mathcal{A}_h(x, y) &:= \begin{pmatrix} \partial g(x) + \nabla f(x) - \nabla h(x) - A^\top y + \sigma A^\top (Ax - b) \\ Ax - b \end{pmatrix}. \end{aligned} \quad (12)$$

The associated PFBS mapping is

$$\widehat{\mathcal{T}}_{\text{lin}}(w) := (\mathcal{M} + \mathcal{A}_h)^{-1}(\mathcal{M}w - \mathcal{B}_h(w)), \quad \mathcal{M} = \begin{pmatrix} P & A^\top \\ A & Q \end{pmatrix}.$$

The following basic facts justify the use of this splitting in a preconditioned forward–backward step.

Proposition 5 *Suppose that h is convex and continuously differentiable, that ∇h is L_h -Lipschitz continuous, and that (11) holds. Then the operator $\mathcal{A}_h$ is maximally monotone, while $\mathcal{B}_h$ is monotone and globally Lipschitz continuous with Lipschitz constant L_h .*

Proof. Define

$$\varphi_h(x) := g(x) + f(x) - h(x) + \frac{\sigma}{2} \|Ax - b\|^2.$$

By (11), the function φ_h is proper, closed, and convex; therefore $\partial\varphi_h$ is maximally monotone. Since

$$\mathcal{A}_h(x, y) = \begin{pmatrix} \partial\varphi_h(x) \\ 0 \end{pmatrix} + \begin{pmatrix} 0 & -A^\top \\ A & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} 0 \\ -b \end{pmatrix},$$

and the block operator is bounded and skew-adjoint, $\mathcal{A}_h$ is maximally monotone. Moreover, $\mathcal{B}_h = \nabla \tilde{h}$ with $\tilde{h}(x, y) := h(x)$; therefore $\mathcal{B}_h$ is monotone. Its Lipschitz bound is immediate from the L_h -Lipschitz continuity of ∇h :

$$\|\mathcal{B}_h(w) - \mathcal{B}_h(w')\| = \|\nabla h(x) - \nabla h(x')\| \leq L_h \|x - x'\| \leq L_h \|w - w'\|.$$

□

Remark 5 (Special choices of the forward part) The two-parameter partial-linearization family is contained in the present framework. If

$$h(x) = \psi_{\theta, \delta}(x) := \theta f(x) + \delta \frac{\sigma}{2} \|Ax - b\|^2, \quad \theta, \delta \in [0, 1],$$

then

$$\nabla h(x) = \theta \nabla f(x) + \delta \sigma A^\top (Ax - b),$$

and (12) reduces to the splitting used in the corresponding partial-linearization scheme. In particular, $h = f$ corresponds to linearizing only the smooth objective term, while $h = \theta f + \delta \frac{\sigma}{2} \|Ax - b\|^2$ yields a family that also partially linearizes the augmented quadratic penalty.

The parameter δ has a concrete computational role. In the primal subproblem, this choice leaves only $(1 - \delta) \frac{\sigma}{2} \|Ax - b\|^2$ in the backward part and treats the remaining δ -fraction by the forward correction $\delta \sigma A^\top (Ax - b)$. Thus $\delta = 0$ keeps the full augmented quadratic implicit, whereas $\delta = 1$ moves it completely to the forward step. Intermediate values provide a tradeoff between retaining useful augmented curvature and reducing the cost of the primal minimization. This can be relevant when $A^\top A$ is dense or poorly structured: keeping the quadratic implicit may require a coupled linear solve or inner iteration, whereas the forward treatment only uses products with A and $A^\top$.

Assumption 2. *Throughout this section,*

$$Q \succ 0, \quad P - A^\top Q^{-1} A \succ 0.$$

Equivalently, $\mathcal{M}$ is positive definite on $\mathbb{R}^n \times \mathbb{R}^m$.

This is a Schur-complement condition. For example, if $Q = \rho^{-1}I_m$ and $P = \tau^{-1}I_n$, it holds whenever $\tau\rho\|A\|^2 < 1$. For the shifted augmented Chambolle–Pock metric $P = \tau^{-1}I_n - \sigma A^\top A$ and $Q = \rho^{-1}I_m$, it is implied by $\tau(\rho + \sigma)\|A\|^2 < 1$.

Together with Proposition 5, this metric assumption ensures that $\widehat{\mathcal{T}}_{\text{lin}}$ is single-valued and everywhere defined. The abstract PFBS map then gives the following explicit primal–dual update.

Proposition 6 *Suppose that h is convex and continuously differentiable, that ∇h is L_h -Lipschitz continuous, and that (11) and Assumption 2 hold. Let $w^k = (x^k, y^k)$ and define $\hat{w}^k = (\hat{x}^k, \hat{y}^k) := \widehat{\mathcal{T}}_{\text{lin}}(w^k)$. Then $(\hat{x}^k, \hat{y}^k)$ is characterized by*

$$\begin{aligned} \hat{x}^k \in \arg \min_x \Big\{ & f(x) + g(x) - h(x) + \langle \nabla h(x^k), x - x^k \rangle - \langle y^k, Ax - b \rangle \\ & + \frac{\sigma}{2} \|Ax - b\|^2 + \frac{1}{2} \|x - x^k\|_P^2 \Big\}, \end{aligned} \quad (13a)$$

$$\hat{y}^k = y^k + Q^{-1}(b - A(2\hat{x}^k - x^k)). \quad (13b)$$

Proof. By definition,

$$\hat{w}^k = (\mathcal{M} + \mathcal{A}_h)^{-1}(\mathcal{M}w^k - \mathcal{B}_h(w^k)) \iff \mathcal{M}w^k - \mathcal{B}_h(w^k) \in \mathcal{M}\hat{w}^k + \mathcal{A}_h(\hat{w}^k).$$

Comparing the two components gives

$$\begin{aligned} Px^k + A^\top y^k - \nabla h(x^k) &\in P\hat{x}^k + \partial g(\hat{x}^k) + \nabla f(\hat{x}^k) - \nabla h(\hat{x}^k) \\ &\quad + \sigma A^\top (A\hat{x}^k - b), \\ Ax^k + Qy^k &= 2A\hat{x}^k + Q\hat{y}^k - b. \end{aligned}$$

That is,

$$0 \in \partial g(\hat{x}^k) + \nabla f(\hat{x}^k) - \nabla h(\hat{x}^k) + \nabla h(x^k) - A^\top y^k + \sigma A^\top (A\hat{x}^k - b) + P(\hat{x}^k - x^k),$$

and

$$Q(\hat{y}^k - y^k) = b - A(2\hat{x}^k - x^k).$$

Now define

$$\begin{aligned} \Xi_k(x) &:= f(x) + g(x) - h(x) + \langle \nabla h(x^k), x - x^k \rangle - \langle y^k, Ax - b \rangle \\ &\quad + \frac{\sigma}{2} \|Ax - b\|^2 + \frac{1}{2} \|x - x^k\|_P^2. \end{aligned}$$

Because $f - h + \frac{\sigma}{2}\|Ax - b\|^2$ is convex by (11), g is proper, closed, and convex, and Assumption 2 implies $P \succ 0$, the function Ξ_k is proper, closed, and strongly convex. Its first-order optimality condition is the displayed primal inclusion above, and is therefore equivalent to (13a); the displayed dual equation is precisely (13b). $\square$

Thus the linearization induced by h replaces $h(x)$ in the primal subproblem by its affine model $h(x^k) + \langle \nabla h(x^k), x - x^k \rangle$; after dropping the irrelevant constant $h(x^k)$, this contributes the term $-h(x) + \langle \nabla h(x^k), x - x^k \rangle$ in (13a).

Under Assumption 2, the operator $\mathcal{M}$ is invertible. We therefore set

$$\mathcal{C} := \mathcal{M}^{-1}\mathcal{B}_h, \quad \mathcal{G} := I - \mathcal{C}, \quad \mathcal{J} := (\mathcal{M} + \mathcal{A}_h)^{-1}\mathcal{M},$$

which gives

$$\widehat{\mathcal{T}}_{\text{lin}} = \mathcal{J} \circ \mathcal{G}.$$

3.2 Averagedness and the Relaxation Range

We use the $\mathcal{M}$ -metric notation $\langle u, v \rangle_{\mathcal{M}} := \langle u, \mathcal{M}v \rangle$ and $\|u\|_{\mathcal{M}} := \sqrt{\langle u, u \rangle_{\mathcal{M}}}$; unqualified norms are Euclidean. Recall that a mapping R is nonexpansive in this metric if $\|Ru - Rv\|_{\mathcal{M}} \leq \|u - v\|_{\mathcal{M}}$ for all u, v , and is α -averaged, with $\alpha \in (0, 1)$, if

$$R = (1 - \alpha)I + \alpha N$$

for some nonexpansive mapping N in the same metric. Firm nonexpansiveness means $1/2$ -averagedness.

The forward operator $\mathcal{B}_h$ only acts on the primal component; the key metric quantity is the amount by which $\mathcal{M}$ controls that component. Define

$$\mu_x := \lambda_{\min}(P - A^\top Q^{-1}A) > 0.$$

Writing $S_x := P - A^\top Q^{-1}A$, a completion of the square gives, for every primal–dual direction (u, v) ,

$$\|(u, v)\|_{\mathcal{M}}^2 = \|u\|_{S_x}^2 + \|Q^{1/2}(v + Q^{-1}Au)\|^2.$$

Consequently,

$$\|(u, v)\|_{\mathcal{M}}^2 \geq \mu_x \|u\|^2. \quad (14)$$

Thus closeness in the $\mathcal{M}$ -metric implies Euclidean closeness of the primal components. This is the bridge that allows the L_h -Lipschitz continuity of ∇h to be expressed as an averagedness estimate for the forward step in the $\mathcal{M}$ -metric.

Proposition 7 *Suppose that h is convex and continuously differentiable, that ∇h is L_h -Lipschitz continuous, and that (11) and Assumption 2 hold, with $\mu_x > L_h/2$. If $L_h > 0$, then $\mathcal{G} = I - \mathcal{C}$ is α_{fw} -averaged in the $\mathcal{M}$ -metric with $\alpha_{\text{fw}} := L_h/(2\mu_x) \in (0, 1)$. If $L_h = 0$, then $\mathcal{G}$ is an isometry in the $\mathcal{M}$ -metric. In all cases, $\mathcal{J} = (\mathcal{M} + \mathcal{A}_h)^{-1}\mathcal{M}$ is firmly nonexpansive in the $\mathcal{M}$ -metric, and $\widehat{\mathcal{T}}_{\text{lin}} = \mathcal{J} \circ \mathcal{G}$ is α_{lin} -averaged, where*

$$\alpha_{\text{lin}} := \frac{1}{2 - \frac{L_h}{2\mu_x}}.$$

Proof. By Proposition 5, $\mathcal{A}_h$ is maximally monotone. Since $\mathcal{M} \succ 0$, the resolvent $\mathcal{J} = (\mathcal{M} + \mathcal{A}_h)^{-1}\mathcal{M}$ is firmly nonexpansive in the $\mathcal{M}$ -metric.

If $L_h = 0$, then ∇h is constant, and therefore $\mathcal{G} = I - \mathcal{M}^{-1}\mathcal{B}_h$ is a translation. Therefore $\|\mathcal{G}w - \mathcal{G}w'\|_{\mathcal{M}} = \|w - w'\|_{\mathcal{M}}$ for all w, w' . Applying the firm nonexpansiveness of $\mathcal{J}$ to the translated inputs $\mathcal{G}w$ and $\mathcal{G}w'$ gives

$$\|\widehat{\mathcal{T}}_{\text{lin}}w - \widehat{\mathcal{T}}_{\text{lin}}w'\|_{\mathcal{M}}^2 \leq \langle \widehat{\mathcal{T}}_{\text{lin}}w - \widehat{\mathcal{T}}_{\text{lin}}w', w - w' \rangle_{\mathcal{M}}$$

because $\mathcal{G}w - \mathcal{G}w' = w - w'$. Thus $\widehat{\mathcal{T}}_{\text{lin}}$ is firmly nonexpansive, i.e., $1/2$ -averaged, which is the value of α_{lin} when $L_h = 0$.

It remains to consider $L_h > 0$. Let $w = (x, y)$, $w' = (x', y')$, and set

$$d := w - w', \quad \xi := \nabla h(x) - \nabla h(x'), \quad c := \mathcal{C}w - \mathcal{C}w'.$$

Since $\mathcal{B}_h(w) - \mathcal{B}_h(w') = (\xi, 0)$ and $\mathcal{M}\mathcal{C} = \mathcal{B}_h$,

$$\langle c, d \rangle_{\mathcal{M}} = \langle \mathcal{B}_h(w) - \mathcal{B}_h(w'), w - w' \rangle = \langle \xi, x - x' \rangle.$$

Because h is convex and ∇h is L_h -Lipschitz continuous, the Baillon–Haddad theorem [4] yields

$$\langle \xi, x - x' \rangle \geq \frac{1}{L_h} \|\xi\|^2.$$

On the other hand, (14) implies $\|z\|_{\mathcal{M}}^2 \geq \mu_x \|u\|^2$ for every $z = (u, v)$. Thus

$$\|(\xi, 0)\|_{\mathcal{M}^{-1}}^2 = \sup_{z \neq 0} \frac{\langle (\xi, 0), z \rangle^2}{\|z\|_{\mathcal{M}}^2} \leq \frac{1}{\mu_x} \|\xi\|^2.$$

Here $\|\cdot\|_{\mathcal{M}^{-1}}$ denotes the norm induced by $\mathcal{M}^{-1}$. Since $\mathcal{M}c = (\xi, 0)$,

$$\|c\|_{\mathcal{M}}^2 = \|\mathcal{M}c\|_{\mathcal{M}^{-1}}^2 \leq \frac{1}{\mu_x} \|\xi\|^2.$$

Combining the two estimates, we obtain

$$\langle c, d \rangle_{\mathcal{M}} \geq \frac{1}{L_h} \|\xi\|^2 \geq \frac{\mu_x}{L_h} \|c\|_{\mathcal{M}}^2.$$

Thus $\mathcal{C}$ is β -cocoercive in the $\mathcal{M}$ -metric with $\beta := \mu_x/L_h$.

Set $q := \mathcal{G}w - \mathcal{G}w' = d - c$. Then

$$\|q\|_{\mathcal{M}}^2 = \|d - c\|_{\mathcal{M}}^2 \leq \|d\|_{\mathcal{M}}^2 - (2\beta - 1)\|c\|_{\mathcal{M}}^2.$$

Since $\mu_x > L_h/2$, one has $\beta > 1/2$, which gives $\alpha_{\text{fw}} = 1/(2\beta) = L_h/(2\mu_x) \in (0, 1)$ and

$$\|\mathcal{G}w - \mathcal{G}w'\|_{\mathcal{M}}^2 \leq \|w - w'\|_{\mathcal{M}}^2 - \frac{1 - \alpha_{\text{fw}}}{\alpha_{\text{fw}}} \|(I - \mathcal{G})w - (I - \mathcal{G})w'\|_{\mathcal{M}}^2,$$

Thus $\mathcal{G}$ is α_{fw} -averaged.

Next let

$$t := \widehat{\mathcal{T}}_{\text{lin}}w - \widehat{\mathcal{T}}_{\text{lin}}w', \quad q := \mathcal{G}w - \mathcal{G}w'.$$

Then $t = \mathcal{J}(\mathcal{G}w) - \mathcal{J}(\mathcal{G}w')$, and firm nonexpansiveness gives

$$\|t\|_{\mathcal{M}}^2 + \|q - t\|_{\mathcal{M}}^2 \leq \|q\|_{\mathcal{M}}^2.$$

Together with the averagedness inequality for $\mathcal{G}$ this yields

$$\|t\|_{\mathcal{M}}^2 + \|q - t\|_{\mathcal{M}}^2 + \frac{1 - \alpha_{\text{fw}}}{\alpha_{\text{fw}}} \|d - q\|_{\mathcal{M}}^2 \leq \|d\|_{\mathcal{M}}^2.$$

Now set $r := d - q$ and $s := q - t$, for which $d - t = r + s$. Writing $\eta := (1 - \alpha_{\text{fw}})/\alpha_{\text{fw}}$, we compute

$$(1 + \eta)(\|s\|_{\mathcal{M}}^2 + \eta\|r\|_{\mathcal{M}}^2) - \eta\|r + s\|_{\mathcal{M}}^2 = \|s - \eta r\|_{\mathcal{M}}^2 \geq 0.$$

Therefore

$$\|s\|_{\mathcal{M}}^2 + \eta\|r\|_{\mathcal{M}}^2 \geq \frac{\eta}{1 + \eta} \|r + s\|_{\mathcal{M}}^2 = (1 - \alpha_{\text{fw}}) \|d - t\|_{\mathcal{M}}^2.$$

Substituting this bound into the previous inequality gives

$$\|t\|_{\mathcal{M}}^2 + (1 - \alpha_{\text{fw}}) \|d - t\|_{\mathcal{M}}^2 \leq \|d\|_{\mathcal{M}}^2.$$

Therefore $\widehat{\mathcal{T}}_{\text{lin}}$ is α_{lin} -averaged with

$$\frac{1 - \alpha_{\text{lin}}}{\alpha_{\text{lin}}} = 1 - \alpha_{\text{fw}}, \quad \text{i.e.,} \quad \alpha_{\text{lin}} = \frac{1}{2 - \alpha_{\text{fw}}} = \frac{1}{2 - \frac{L_h}{2\mu_x}}.$$

□

The averagedness constant immediately determines a sufficient admissible range for the reflected relaxation parameter. We do not use, or claim, a converse necessity statement for this range.

Theorem 2 *Under the assumptions of Proposition 7, define $\mathcal{S}_\gamma := (1 + \gamma)\widehat{\mathcal{T}}_{\text{lin}} - \gamma I$. Then $\mathcal{S}_\gamma$ is nonexpansive in the $\mathcal{M}$ -metric whenever*

$$-1 < \gamma \leq 1 - \frac{L_h}{2\mu_x}.$$

Proof. By Proposition 7, $\widehat{\mathcal{T}}_{\text{lin}} = (1 - \alpha_{\text{lin}})I + \alpha_{\text{lin}}\mathcal{N}$ for some nonexpansive map $\mathcal{N}$ in the $\mathcal{M}$ -metric. Therefore

$$\mathcal{S}_\gamma = (1 - (1 + \gamma)\alpha_{\text{lin}})I + (1 + \gamma)\alpha_{\text{lin}}\mathcal{N}.$$

Thus $\mathcal{S}_\gamma$ is nonexpansive whenever $(1 + \gamma)\alpha_{\text{lin}} \leq 1$ and the coefficients are nonnegative. Since $\alpha_{\text{lin}} > 0$, these two requirements are equivalent to

$$-1 < \gamma \leq \alpha_{\text{lin}}^{-1} - 1 = 1 - \frac{L_h}{2\mu_x},$$

which is exactly the claimed range. □

For the two main choices used later, the following Schur complements and convenient sufficient conditions are obtained.

Corollary 1 *In the special case $h = f$, where $L_h = L_f$, the following statements hold:*

(i) *if $P = \tau^{-1}I - \sigma A^\top A$ and $Q = \rho^{-1}I$, then*

$$\mu_x = \lambda_{\min}(\tau^{-1}I - (\sigma + \rho)A^\top A),$$

and the threshold $\mu_x > L_f/2$ is implied by

$$\tau\left((\sigma + \rho)\|A\|^2 + \frac{L_f}{2}\right) < 1;$$

(ii) *if $P = \tau^{-1}I$ and $Q = \rho^{-1}I$, then*

$$\mu_x = \lambda_{\min}(\tau^{-1}I - \rho A^\top A),$$

and the threshold $\mu_x > L_f/2$ is implied by

$$\tau\left(\rho\|A\|^2 + \frac{L_f}{2}\right) < 1.$$

Proof. When $h = f$, we have $L_h = L_f$. In case (i), $Q^{-1} = \rho I$ and $P - A^\top Q^{-1}A = \tau^{-1}I - (\sigma + \rho)A^\top A$. Therefore

$$\mu_x \geq \tau^{-1} - (\sigma + \rho)\|A\|^2,$$

and the displayed condition in (i) implies $\mu_x > L_f/2$. Case (ii) is identical, with $P - A^\top Q^{-1}A = \tau^{-1}I - \rho A^\top A$, and the displayed condition in (ii) again implies $\mu_x > L_f/2$. $\square$

Remark 6 A direct curvature-compensation argument yields the more conservative condition $\mu_x \geq L_h$. The operator-theoretic PFBS analysis shows that averagedness already holds under the weaker threshold $\mu_x > L_h/2$. In particular, the admissible relaxation range is governed by the Schur complement quantity $\mu_x = \lambda_{\min}(P - A^\top Q^{-1}A)$.

3.3 Reflected Halpern PFBS and Nonergodic KKT Rates

We now pass from the map-level statement of Theorem 2 to the reflected Halpern PFBS iteration, with relaxation parameter restricted to $\gamma \in (-1, 1 - L_h/(2\mu_x)]$. Using Proposition 6, the PFBS shadow step can be written explicitly in primal-dual variables. Algorithm 2 is the linearized counterpart of Algorithm 1: the first two steps compute the PFBS shadow point, and the last two steps apply the same reflection and Halpern anchoring.

Algorithm 2: Reflected Halpern-Accelerated Linearized PFBS

1: **Input:** $w^0 = (x^0, y^0)$ and $\gamma \in (-1, 1 - L_h/(2\mu_x)]$.
 2: **for** $k = 0, 1, 2, \dots$ **do**
 3: **Step 1.** Compute

$$\hat{x}^k = \arg \min_x \left\{ \begin{aligned} & f(x) + g(x) - h(x) + \langle \nabla h(x^k), x - x^k \rangle - \langle y^k, Ax - b \rangle \\ & + \frac{\sigma}{2} \|Ax - b\|^2 + \frac{1}{2} \|x - x^k\|_P^2 \end{aligned} \right\}$$

4: **Step 2.** Set $\hat{y}^k = y^k + Q^{-1}(b - A(2\hat{x}^k - x^k))$ and $\hat{w}^k = (\hat{x}^k, \hat{y}^k)$.
 5: **Step 3.** $\bar{w}^k = (1 + \gamma)\hat{w}^k - \gamma w^k$.
 6: **Step 4.** $w^{k+1} = (w^0 + (k+1)\bar{w}^k)/(k+2)$.
 7: **end for**
 8: **Output:** shadow sequence $\{\hat{w}^k\}$.

The following proposition records convergence and fixed-point residual estimates for this reflected Halpern PFBS scheme.

Proposition 8 *Assume that h is convex and continuously differentiable, that ∇h is L_h -Lipschitz continuous, and that (11), Assumption 2, $\mu_x > L_h/2$, and $\mathcal{T}^{-1}(0) \neq \emptyset$. Fix $\gamma \in (-1, 1 - L_h/(2\mu_x)]$ and define $\mathcal{S}_\gamma := (1 + \gamma)\hat{\mathcal{T}}_{\text{lin}} - \gamma I$. Let the sequences $\{w^k\}$, $\{\hat{w}^k\}$, and $\{\bar{w}^k\}$ be generated by Algorithm 2. Then there exists $w^* \in \mathcal{T}^{-1}(0)$ such that*

$$w^k \rightarrow w^*, \quad \bar{w}^k \rightarrow w^*, \quad \hat{w}^k \rightarrow w^*, \quad (15)$$

and, for every $k \geq 0$,

$$\begin{aligned} \|w^k - \bar{w}^k\|_{\mathcal{M}} &\leq \frac{2}{k+1} \text{dist}_{\mathcal{M}}(w^0, \mathcal{T}^{-1}(0)), \\ \|w^k - \hat{w}^k\|_{\mathcal{M}} &\leq \frac{2}{(1+\gamma)(k+1)} \text{dist}_{\mathcal{M}}(w^0, \mathcal{T}^{-1}(0)). \end{aligned} \quad (16)$$

Proof. By Theorem 2, $\mathcal{S}_\gamma$ is nonexpansive in the $\mathcal{M}$ -metric. Here and below, $\text{Fix}(R) := \{w : Rw = w\}$ denotes the fixed-point set of a map R . Since $1 + \gamma > 0$,

$$w = \mathcal{S}_\gamma(w) \iff (1 + \gamma)(\hat{\mathcal{T}}_{\text{lin}}(w) - w) = 0 \iff w = \hat{\mathcal{T}}_{\text{lin}}(w),$$

which gives $\text{Fix}(\mathcal{S}_\gamma) = \text{Fix}(\hat{\mathcal{T}}_{\text{lin}})$. By the definition of $\hat{\mathcal{T}}_{\text{lin}}$,

$$\begin{aligned} w = \hat{\mathcal{T}}_{\text{lin}}(w) &\iff \mathcal{M}w - \mathcal{B}_h(w) \in \mathcal{M}w + \mathcal{A}_h(w) \\ &\iff 0 \in (\mathcal{A}_h + \mathcal{B}_h)(w) = \mathcal{T}(w), \end{aligned}$$

therefore $\text{Fix}(\mathcal{S}_\gamma) = \mathcal{T}^{-1}(0)$.

Algorithm 2 is precisely the Halpern iteration associated with the nonexpansive map $\mathcal{S}_\gamma$ in the Hilbert space $(\mathbb{R}^n \times \mathbb{R}^m, \langle \cdot, \cdot \rangle_{\mathcal{M}})$, with anchor w^0 and weights $1/(k+2)$. The Halpern convergence theorem for nonexpansive mappings therefore yields $w^k \rightarrow w^*$ for some $w^* \in \mathcal{T}^{-1}(0)$; see, e.g., [18, 23]. Continuity of $\mathcal{S}_\gamma$ gives $\bar{w}^k = \mathcal{S}_\gamma(w^k) \rightarrow w^*$, and

$$\hat{w}^k = \frac{1}{1+\gamma}(\bar{w}^k + \gamma w^k)$$

then implies $\hat{w}^k \rightarrow w^*$, proving (15).

The corresponding Halpern residual estimate [23] gives

$$\|w^k - \bar{w}^k\|_{\mathcal{M}} \leq \frac{2}{k+1} \text{dist}_{\mathcal{M}}(w^0, \mathcal{T}^{-1}(0)).$$

Since $w^k - \bar{w}^k = (1 + \gamma)(w^k - \hat{w}^k)$, the second estimate in (16) follows immediately. $\square$

We now translate the fixed-point residual estimate into the KKT residual and the objective-value gap of the original constrained problem. Set $\Phi := f + g$, and recall from (10) the original KKT mapping $\mathcal{R}$ and residual r_{KKT} . For $w = (x, y)$, we also write $r_{\text{KKT}}(w) := r_{\text{KKT}}(x, y)$. Let $\mathcal{E}_\sigma$ denote the linear map $\mathcal{E}_\sigma(u, v) := (u - \sigma A^\top v, v)$.

Theorem 3 *Under the assumptions of Proposition 8, let $w^* = (x^*, y^*)$ be the common limit from (15). Then $w^* \in \mathcal{T}^{-1}(0) = \mathcal{R}^{-1}(0)$ and, for every $k \geq 0$,*

$$r_{\text{KKT}}(\hat{w}^k) \leq \frac{2\|\mathcal{E}_\sigma\|}{(1 + \gamma)(k + 1)} \left(\sqrt{\|\mathcal{M}\|} + \frac{L_h}{\sqrt{\mu_x}} \right) \text{dist}_{\mathcal{M}}(w^0, \mathcal{T}^{-1}(0)),$$

in particular $r_{\text{KKT}}(\hat{w}^k) = O(1/k)$. Set

$$C_* := \sup_{k \geq 0} (\|\hat{x}^k - x^*\| + \|\hat{y}^k\|) + \|y^*\|.$$

Then $C_* < \infty$ and

$$|\Phi(\hat{x}^k) - \Phi(x^*)| \leq C_* r_{\text{KKT}}(\hat{w}^k),$$

and consequently $|\Phi(\hat{x}^k) - \Phi(x^*)| = O(1/k)$.

Proof. By Proposition 8, $w^k \rightarrow w^*$ and $\hat{w}^k \rightarrow w^*$ for some $w^* \in \mathcal{T}^{-1}(0)$. Since the second component of $\mathcal{T}$ is $Ax - b$, the augmentation term vanishes on $\mathcal{T}^{-1}(0)$; therefore $\mathcal{T}^{-1}(0) = \mathcal{R}^{-1}(0)$.

From the definition of $\hat{\mathcal{T}}_{\text{lin}}$,

$$\hat{w}^k = (\mathcal{M} + \mathcal{A}_h)^{-1}(\mathcal{M}w^k - \mathcal{B}_h(w^k)),$$

and therefore

$$\mathcal{M}w^k - \mathcal{B}_h(w^k) \in \mathcal{M}\hat{w}^k + \mathcal{A}_h(\hat{w}^k).$$

It follows that

$$d^k := \mathcal{M}(w^k - \hat{w}^k) + \mathcal{B}_h(\hat{w}^k) - \mathcal{B}_h(w^k) \in \mathcal{T}(\hat{w}^k),$$

and therefore $\text{dist}(0, \mathcal{T}(\hat{w}^k)) \leq \|d^k\|$. Since

$$\|\mathcal{B}_h(\hat{w}^k) - \mathcal{B}_h(w^k)\| = \|\nabla h(\hat{x}^k) - \nabla h(x^k)\| \leq L_h \|\hat{x}^k - x^k\| \leq \frac{L_h}{\sqrt{\mu_x}} \|w^k - \hat{w}^k\|_{\mathcal{M}},$$

where the last step uses (14), and

$$\|\mathcal{M}(w^k - \hat{w}^k)\| \leq \sqrt{\|\mathcal{M}\|} \|w^k - \hat{w}^k\|_{\mathcal{M}},$$

we obtain

$$\text{dist}(0, \mathcal{T}(\hat{w}^k)) \leq \left(\sqrt{\|\mathcal{M}\|} + \frac{L_h}{\sqrt{\mu_x}} \right) \|w^k - \hat{w}^k\|_{\mathcal{M}}.$$

Combining this with (16) yields

$$\text{dist}(0, \mathcal{T}(\hat{w}^k)) \leq \frac{2}{(1+\gamma)(k+1)} \left(\sqrt{\|\mathcal{M}\|} + \frac{L_h}{\sqrt{\mu_x}} \right) \text{dist}_{\mathcal{M}}(w^0, \mathcal{T}^{-1}(0)).$$

Now, for any (x, y) and any $(u, v) \in \mathcal{T}(x, y)$,

$$u \in \nabla f(x) + \partial g(x) - A^\top y + \sigma A^\top (Ax - b), \quad v = Ax - b,$$

we have $\mathcal{E}_\sigma(u, v) \in \mathcal{R}(x, y)$. Therefore

$$r_{\text{KKT}}(\hat{w}^k) \leq \|\mathcal{E}_\sigma\| \text{dist}(0, \mathcal{T}(\hat{w}^k)),$$

which proves the displayed residual estimate.

For the objective-value bound, the preceding inclusion implies $\mathcal{R}(\hat{w}^k) \neq \emptyset$. This set is closed and convex in finite dimensions; choose a least-norm element $r^k = (r_x^k, r_y^k) \in \mathcal{R}(\hat{w}^k)$. Then $\|r^k\| = r_{\text{KKT}}(\hat{w}^k)$, $r_y^k = A\hat{x}^k - b$ and $r_x^k \in \partial\Phi(\hat{x}^k) - A^\top \hat{y}^k$. Thus there exists $\zeta^k \in \partial\Phi(\hat{x}^k)$ with $\zeta^k = A^\top \hat{y}^k + r_x^k$. By convexity of Φ ,

$$\begin{aligned} \Phi(\hat{x}^k) - \Phi(x^*) &\leq \langle \zeta^k, \hat{x}^k - x^* \rangle \\ &= \langle r_x^k, \hat{x}^k - x^* \rangle + \langle \hat{y}^k, A\hat{x}^k - b \rangle \\ &\leq (\|\hat{x}^k - x^*\| + \|\hat{y}^k\|) r_{\text{KKT}}(\hat{w}^k). \end{aligned}$$

Likewise, since $w^* \in \mathcal{R}^{-1}(0)$, there exists $\zeta^* \in \partial\Phi(x^*)$ such that $\zeta^* = A^\top y^*$, and convexity gives

$$\Phi(\hat{x}^k) - \Phi(x^*) \geq \langle \zeta^*, \hat{x}^k - x^* \rangle = \langle y^*, A\hat{x}^k - b \rangle \geq -\|y^*\| r_{\text{KKT}}(\hat{w}^k).$$

Since $\hat{w}^k \rightarrow w^*$, the constant C_* is finite. Combining the upper and lower bounds yields the displayed objective-value estimate, and the $O(1/k)$ conclusion follows from the residual estimate. $\square$

Remark 7 (Worst-case tightness) The nonergodic residual bound in Theorem 3 is worst-case tight even within the linearized PFBS family. Fix $\gamma \in (-1, 1)$ and choose $\eta \in (0, 1)$ such that $(1+\gamma)\eta < 1$ and $\gamma \leq 1 - \eta/2$. Consider the scalar instance with $g = 0$, $A = 0$, $b = 0$, $\sigma = 0$, $P = 1$, $Q = 1$, and

$$f(x) = h(x) = \frac{\eta}{2} x^2.$$

Then $L_h = \eta$, $\mu_x = 1$, and the admissible relaxation condition is satisfied. The PFBS shadow map is $F(x) = \hat{\mathcal{T}}_{\text{lin}}(x) = (1 - \eta)x$; the dual coordinate remains

zero if initialized at zero. Starting from $x^0 \neq 0$, the reflected Halpern recursion gives, with $q := 1 - (1 + \gamma)\eta$,

$$x^k = \frac{1 - q^{k+1}}{(1 + \gamma)\eta(k + 1)} x^0, \quad \hat{x}^k = (1 - \eta)x^k.$$

The KKT residual at the shadow point is $r_{\text{KKT}}(\hat{x}^k, 0) = \eta|\hat{x}^k|$. Therefore

$$\lim_{k \rightarrow \infty} (k + 1)r_{\text{KKT}}(\hat{x}^k, 0) = \frac{(1 - \eta)|x^0|}{1 + \gamma} > 0.$$

Thus the global nonergodic KKT-residual estimate cannot, in general, be improved to $o(1/k)$.

4 Shadow Identification and Reduced Geometry for Reflected Halpern Schemes

For reflected Halpern schemes, the relevant local object is the shadow point, not the anchored Halpern state. Let $W^* := \mathcal{T}^{-1}(0)$ denote the KKT set. For $F \in \{\widehat{\mathcal{T}}, \widehat{\mathcal{T}}_{\text{lin}}\}$, a reflected Halpern step has the form

$$\widehat{w}^k = F(w^k), \quad \bar{w}^k = (1 + \gamma)\widehat{w}^k - \gamma w^k, \quad w^{k+1} = \frac{1}{k + 2}w^0 + \frac{k + 1}{k + 2}\bar{w}^k.$$

Thus w^{k+1} is obtained by anchoring and reflecting the shadow $\widehat{w}^k$. Even when $\widehat{x}^k$ lies on the active manifold, the state x^{k+1} need not lie on that manifold.

The local analysis is therefore carried out at the shadow points. We prove finite identification of $\widehat{x}^k$ for both fixed-point maps, derive an exact reduced residual identity under an affine-face model, and describe the perturbed reduced dynamics retained by the anchored Halpern state.

Throughout this section, all norms and distances are Euclidean unless another metric is explicitly specified.

4.1 Shadow Certificates and Finite Identification

We first record that the augmented term changes the geometry of the operator, but not the solution set. Recall from (10) the original KKT mapping $\mathcal{R}$, and define the augmented correction by

$$\mathcal{C}_\sigma(x, y) := \begin{pmatrix} \sigma A^\top (Ax - b) \\ 0 \end{pmatrix}.$$

Then $\mathcal{T} = \mathcal{R} + \mathcal{C}_\sigma$.

Proposition 9 *One has $\mathcal{T}^{-1}(0) = \mathcal{R}^{-1}(0) = W^*$. Moreover, for every $w = (x, y) \in \mathbb{R}^n \times \mathbb{R}^m$ and every $w^* = (x^*, y^*) \in W^*$,*

$$\langle \mathcal{C}_\sigma(w), w - w^* \rangle = \sigma \|Ax - b\|^2.$$

Proof. Since $\mathcal{T} = \mathcal{R} + \mathcal{C}_\sigma$, and since the common second component $Ax - b$ forces $\mathcal{C}_\sigma(x, y) = 0$ at every zero of either operator, their zero sets coincide. If $w^\star = (x^\star, y^\star) \in W^\star$, then $Ax^\star = b$, and therefore

$$\langle \mathcal{C}_\sigma(w), w - w^\star \rangle = \sigma \langle A^\top (Ax - b), x - x^\star \rangle = \sigma \|Ax - b\|^2.$$

□

We now derive the subgradient certificates generated by the shadow point. The statement is formulated simultaneously for the exact map and the linearized map. For $F \in \{\widehat{\mathcal{T}}, \widehat{\mathcal{T}}_{\text{lin}}\}$ and $\widehat{w} = F(w)$, write $w = (x, y)$ and $\widehat{w} = (\widehat{x}, \widehat{y})$. Define the correction term

$$\Delta_F(w, \widehat{w}) := \begin{cases} 0, & F = \widehat{\mathcal{T}}, \\ \nabla h(\widehat{x}) - \nabla h(x), & F = \widehat{\mathcal{T}}_{\text{lin}}. \end{cases}$$

Proposition 10 *Let $F \in \{\widehat{\mathcal{T}}, \widehat{\mathcal{T}}_{\text{lin}}\}$, let $w = (x, y)$, and let $\widehat{w} = (\widehat{x}, \widehat{y}) = F(w)$. Then there exists $v_F(w) \in \partial g(\widehat{x})$ such that*

$$v_F(w) = A^\top y - \nabla f(\widehat{x}) - \sigma A^\top (A\widehat{x} - b) - P(\widehat{x} - x) + \Delta_F(w, \widehat{w}). \quad (17)$$

Consequently, if $w^j \rightarrow w^\star = (x^\star, y^\star) \in W^\star$ and $\widehat{w}^j = F(w^j) \rightarrow w^\star$, then

$$v_F(w^j) \rightarrow v^\star := A^\top y^\star - \nabla f(x^\star) \in \partial g(x^\star). \quad (18)$$

Proof. We first consider the exact dPPM map. Since

$$\widehat{w} = (\mathcal{M} + \mathcal{T})^{-1} \mathcal{M}w,$$

we have

$$\mathcal{M}w \in \mathcal{M}\widehat{w} + \mathcal{T}(\widehat{w}).$$

Comparing the primal component gives

$$0 \in \nabla f(\widehat{x}) + \partial g(\widehat{x}) - A^\top y + \sigma A^\top (A\widehat{x} - b) + P(\widehat{x} - x).$$

Thus there exists $v_F(w) \in \partial g(\widehat{x})$ satisfying

$$v_F(w) = A^\top y - \nabla f(\widehat{x}) - \sigma A^\top (A\widehat{x} - b) - P(\widehat{x} - x),$$

which is exactly (17) with $\Delta_F = 0$.

For the linearized PFBS map, the defining relation is

$$(\mathcal{M} - \mathcal{B}_h)w \in (\mathcal{M} + \mathcal{A}_h)\widehat{w}.$$

Equivalently, the primal optimality condition reads

$$0 \in \partial g(\widehat{x}) + \nabla f(\widehat{x}) - \nabla h(\widehat{x}) + \nabla h(x) - A^\top y + \sigma A^\top (A\widehat{x} - b) + P(\widehat{x} - x).$$

Therefore there exists $v_F(w) \in \partial g(\widehat{x})$ such that

$$v_F(w) = A^\top y - \nabla f(\widehat{x}) + \nabla h(\widehat{x}) - \nabla h(x) - \sigma A^\top (A\widehat{x} - b) - P(\widehat{x} - x),$$

which is again (17).

Now assume $w^j \rightarrow w^* \in W^*$ and $\hat{w}^j \rightarrow w^*$. Then

$$\hat{x}^j \rightarrow x^*, \quad y^j \rightarrow y^*, \quad A\hat{x}^j - b \rightarrow 0, \quad \hat{x}^j - x^j \rightarrow 0.$$

In the linearized case, continuity of ∇h gives $\nabla h(\hat{x}^j) - \nabla h(x^j) \rightarrow 0$. Passing to the limit in (17) yields $v_F(w^j) \rightarrow A^\top y^* - \nabla f(x^*)$. Since $w^* \in W^*$, the KKT condition gives $A^\top y^* - \nabla f(x^*) \in \partial g(x^*)$. This proves (18). $\square$

We use the standard notion of partial smoothness for convex functions in the sense of Lewis [21], with the equivalent tangent-space form of normal sharpness in the convex setting. Here a C^2 -manifold means an embedded manifold with twice continuously differentiable local charts; $T_{\mathcal{M}}(x)$ denotes the tangent space to a manifold $\mathcal{M}$ at x , $\text{par } S := \text{span}(S - S)$ denotes the parallel subspace of a convex set S , and $\text{ri } S$ denotes its relative interior. Let g be proper, closed, and convex, and let $x^* \in \text{dom } g$. We say that g is partly smooth at x^* relative to a C^2 -manifold $\mathcal{M}_g(x^*)$ if, locally around x^* , the restriction of g to $\mathcal{M}_g(x^*)$ is C^2 , the sharpness relation

$$T_{\mathcal{M}_g(x^*)}(x^*) = \text{par}(\partial g(x^*))^\perp$$

holds, and the subdifferential mapping ∂g is continuous at x^* along $\mathcal{M}_g(x^*)$ in the Painlevé–Kuratowski sense. The manifold $\mathcal{M}_g(x^*)$ is then called the active manifold of g at x^* . The following assumption is the corresponding standard identifiability condition at the limiting KKT point; see, e.g., [19].

Assumption 3 (Partial smoothness and nondegeneracy). *Let $w^* = (x^*, y^*) \in W^*$. Assume that*

- (i) *g is proper, closed, convex, and partly smooth at x^* relative to a C^2 -manifold $\mathcal{M}_g(x^*)$;*
- (ii) *the nondegeneracy condition $A^\top y^* - \nabla f(x^*) \in \text{ri } \partial g(x^*)$ holds.*

This assumption is local at the limiting KKT point. It holds for many standard regularizers and constraints under the usual strict-complementarity condition: for instance, for an indicator of a polyhedron or a box, the manifold is the identified face and the relative-interior condition is the standard nondegeneracy condition on the active multipliers.

The next theorem applies the standard identification principle for partly smooth convex functions to the shadow sequence.

Theorem 4 *Let $F \in \{\widehat{T}, \widehat{T}_{\text{lin}}\}$. Let $\{w^j\}_{j \geq 0}$ be any sequence and define the shadow sequence*

$$\hat{w}^j = F(w^j).$$

Assume that

$$w^j \rightarrow w^*, \quad \hat{w}^j \rightarrow w^* \tag{19}$$

for some $w^ = (x^*, y^*) \in W^*$. If Assumption 3 holds at w^* , then there exists an integer $J_{\text{id}} \geq 0$ such that*

$$\hat{x}^j \in \mathcal{M}_g(x^*), \quad \forall j \geq J_{\text{id}}.$$

Proof. By Proposition 10, for every j there exists

$$v_F(w^j) \in \partial g(\hat{x}^j)$$

such that

$$v_F(w^j) \rightarrow v^* := A^\top y^* - \nabla f(x^*).$$

Moreover, (19) gives

$$\hat{x}^j \rightarrow x^*.$$

By Assumption 3(ii),

$$v^* \in \text{ri } \partial g(x^*).$$

Thus the graph points

$$(\hat{x}^j, v_F(w^j)) \in \text{gph } \partial g$$

converge to

$$(x^*, v^*), \quad v^* \in \text{ri } \partial g(x^*).$$

The standard identification theorem for partly smooth convex functions [21, 19] then implies that

$$\hat{x}^j \in \mathcal{M}_g(x^*)$$

for all sufficiently large j . $\square$

Remark 8 (Application to reflected and restarted Halpern iterates) Theorem 4 is intentionally stated for an arbitrary sequence $\{w^j\}$. For the reflected Halpern schemes of Sections 2 and 3, the convergence condition (19) is supplied by the global convergence results for the corresponding nonexpansive reflected maps. For restarted schemes, the same theorem applies to any globally indexed restarted trajectory whenever the state and shadow sequences converge to the same KKT point. The conclusion is always identification of the shadow points $\hat{x}^j$, not identification of the Halpern states x^j .

Remark 9 (The Halpern state need not identify in finite time) Finite identification of the Halpern state x^j is false in general. Consider the one-dimensional problem with no linear constraint,

$$f \equiv 0, \quad g = \delta_{\{0\}}.$$

The active manifold is $\mathcal{M}_g = \{0\}$, and the proximal map satisfies $F(x) = 0$ for every x . For the ordinary Halpern choice $\gamma = 0$,

$$\hat{x}^k = 0, \quad \bar{x}^k = 0, \quad x^{k+1} = \frac{1}{k+2}x^0.$$

Thus $\hat{x}^k \in \mathcal{M}_g$ for every k , but if $x^0 \neq 0$, then $x^k \notin \mathcal{M}_g$ for every finite k . This shows that the shadow sequence is the correct identification object for Halpern-type methods.

4.2 Reduced Residuals on the Identified Affine Face

Partial smoothness yields finite identification of a smooth active manifold. To obtain an exact reduced residual identity, we impose the following stronger local model: the identified manifold is locally an affine face and the subdifferential has a full normal-space fiber on that face. This is a sufficient model for the exact identity below, not a standing assumption for all structured nonsmooth terms.

Assumption 4 (Affine-face model). *Let $w^* = (x^*, y^*) \in W^*$. Assume that there exist an affine set $\mathcal{F} \subset \mathbb{R}^n$ with $x^* \in \mathcal{F}$, a neighborhood U_x of x^* , a symmetric matrix $H_{\mathcal{F}} \succeq 0$, and a vector $c_{\mathcal{F}} \in \mathbb{R}^n$ such that, writing*

$$\mathcal{L}_{\mathcal{F}} := \text{span}(\mathcal{F} - \mathcal{F}), \quad \Pi_{\mathcal{F}} : \mathbb{R}^n \rightarrow \mathcal{L}_{\mathcal{F}}$$

for the tangent space and its orthogonal projector, one has

$$\partial g(x) = H_{\mathcal{F}}x + c_{\mathcal{F}} + \mathcal{L}_{\mathcal{F}}^{\perp}, \quad \forall x \in \mathcal{F} \cap U_x.$$

Assumption 4 is satisfied, for example, by functions of the form $g(x) = \frac{1}{2}\langle x, Hx \rangle + \langle c, x \rangle + \delta_{\mathcal{F}}(x)$ on an affine set $\mathcal{F}$, and more generally by piecewise linear-quadratic terms on a fixed affine region whose subdifferential fiber is an affine translate of the full normal space. It is stronger than partial smoothness and is used only for the exact reduced residual identity below. For box and other polyhedral inequality constraints, the error-bound arguments in Section 5.3 provide the corresponding sharpness verification without requiring this exact affine-face identity.

For $u \in \mathcal{L}_{\mathcal{F}}$, define the affine embedding

$$\Psi(u, y) := (x^* + u, y),$$

and, for $(u, y) \in \mathcal{L}_{\mathcal{F}} \times \mathbb{R}^m$, define the reduced KKT mapping

$$\mathcal{T}_{\text{red}}(u, y) := \begin{pmatrix} \Pi_{\mathcal{F}} \left(\nabla f(x^* + u) + H_{\mathcal{F}}(x^* + u) + c_{\mathcal{F}} \right) \\ - A^{\top} y + \sigma A^{\top} Au \\ Au \end{pmatrix}. \quad (20)$$

Here we used $Ax^* = b$, which gives

$$A(x^* + u) - b = Au.$$

Under the affine-face model, the full augmented KKT residual at points on the face has an exact reduced representation.

Theorem 5 *Suppose Assumption 4 holds. Then there exists a neighborhood U_{red} of $z^* := (0, y^*)$ in $\mathcal{L}_{\mathcal{F}} \times \mathbb{R}^m$ such that, for every $z = (u, y) \in U_{\text{red}}$,*

$$\text{dist}(0, \mathcal{T}(\Psi(z))) = \|\mathcal{T}_{\text{red}}(z)\|. \quad (21)$$

Proof. By Assumption 4, after shrinking U_{red} if necessary, every $z = (u, y) \in U_{\text{red}}$ satisfies $x := x^* + u \in \mathcal{F} \cap U_x$. Define

$$a(z) := \nabla f(x^* + u) + H_{\mathcal{F}}(x^* + u) + c_{\mathcal{F}} - A^{\top}y + \sigma A^{\top}Au.$$

Since $x = x^* + u \in \mathcal{F}$, the affine-face representation gives

$$\mathcal{T}(\Psi(z)) = \begin{pmatrix} a(z) + \mathcal{L}_{\mathcal{F}}^{\perp} \\ Au \end{pmatrix}, \quad \mathcal{T}_{\text{red}}(z) = \begin{pmatrix} \Pi_{\mathcal{F}}a(z) \\ Au \end{pmatrix}.$$

For any $n \in \mathcal{L}_{\mathcal{F}}^{\perp}$,

$$\|a(z) + n\|^2 = \|\Pi_{\mathcal{F}}a(z)\|^2 + \|(I - \Pi_{\mathcal{F}})a(z) + n\|^2 \geq \|\Pi_{\mathcal{F}}a(z)\|^2.$$

Equality is attained at $n = -(I - \Pi_{\mathcal{F}})a(z) \in \mathcal{L}_{\mathcal{F}}^{\perp}$. Therefore $\text{dist}(0, a(z) + \mathcal{L}_{\mathcal{F}}^{\perp}) = \|\Pi_{\mathcal{F}}a(z)\|$. Combining this identity with the product structure of $\mathcal{T}(\Psi(z))$ yields

$$\text{dist}(0, \mathcal{T}(\Psi(z)))^2 = \|\Pi_{\mathcal{F}}a(z)\|^2 + \|Au\|^2 = \|\mathcal{T}_{\text{red}}(z)\|^2,$$

which proves (21). $\square$

Corollary 2 *Let U_{red} be as in Theorem 5, and set*

$$Z_{\text{red}} := U_{\text{red}} \cap \mathcal{T}_{\text{red}}^{-1}(0).$$

Then $z^ \in Z_{\text{red}}$, and therefore Z_{red} is nonempty. Moreover,*

$$z \in Z_{\text{red}} \implies \Psi(z) \in W^*,$$

and

$$\text{dist}(\Psi(z), W^*) \leq \text{dist}(z, Z_{\text{red}}), \quad \forall z \in U_{\text{red}}. \quad (22)$$

Consequently, if there exist a neighborhood $V_{\text{red}} \subset U_{\text{red}}$ of z^ in $\mathcal{L}_{\mathcal{F}} \times \mathbb{R}^m$ and a constant $\alpha_{\text{red}} > 0$ such that*

$$\alpha_{\text{red}} \text{dist}(z, Z_{\text{red}}) \leq \|\mathcal{T}_{\text{red}}(z)\|, \quad \forall z \in V_{\text{red}}, \quad (23)$$

then

$$\alpha_{\text{red}} \text{dist}(w, W^*) \leq \text{dist}(0, \mathcal{T}(w)), \quad \forall w \in \Psi(V_{\text{red}}).$$

Proof. For $z = (u, y) \in U_{\text{red}}$, set

$$a(z) := \nabla f(x^* + u) + H_{\mathcal{F}}(x^* + u) + c_{\mathcal{F}} - A^{\top}y + \sigma A^{\top}Au.$$

Because $w^* \in W^*$ and $x^* \in \mathcal{F} \cap U_x$, the affine-face model and the KKT condition imply

$$a(z^*) = \nabla f(x^*) + H_{\mathcal{F}}x^* + c_{\mathcal{F}} - A^{\top}y^* \in \mathcal{L}_{\mathcal{F}}^{\perp}.$$

The second component of $\mathcal{T}_{\text{red}}(z^*)$ is also zero. Thus $\mathcal{T}_{\text{red}}(z^*) = 0$, and $z^* \in Z_{\text{red}}$.

Suppose $z \in Z_{\text{red}}$. Then

$$\Pi_{\mathcal{F}} a(z) = 0, \quad Au = 0.$$

The first condition implies $a(z) \in \mathcal{L}_{\mathcal{F}}^{\perp}$, and therefore

$$0 \in a(z) + \mathcal{L}_{\mathcal{F}}^{\perp}.$$

Together with $Au = 0$, this gives $0 \in \mathcal{T}(\Psi(z))$, and therefore $\Psi(z) \in W^*$.

Because Ψ is an isometry from $\mathcal{L}_{\mathcal{F}} \times \mathbb{R}^m$ into $\mathbb{R}^n \times \mathbb{R}^m$, taking the infimum over $\bar{z} \in Z_{\text{red}}$ gives

$$\text{dist}(\Psi(z), W^*) \leq \text{dist}(z, Z_{\text{red}}),$$

which proves (22).

Finally, if (23) holds and $w = \Psi(z)$ with $z \in V_{\text{red}}$, then by (22), (23), and (21),

$$\alpha_{\text{red}} \text{dist}(w, W^*) \leq \alpha_{\text{red}} \text{dist}(z, Z_{\text{red}}) \leq \|\mathcal{T}_{\text{red}}(z)\| = \text{dist}(0, \mathcal{T}(w)).$$

This proves the claimed face-restricted error bound. $\square$

The reduced error bound (23) may hold even when the local solution set is not isolated. The following Jacobian condition is a simple sufficient condition for the isolated case.

Proposition 11 *Suppose Assumption 4 holds and f is twice continuously differentiable near x^* . Then $\mathcal{T}_{\text{red}}$ is continuously differentiable near $z^* = (0, y^*)$, with Jacobian*

$$D\mathcal{T}_{\text{red}}(z^*) = \begin{pmatrix} \Pi_{\mathcal{F}}(\nabla^2 f(x^*) + H_{\mathcal{F}} + \sigma A^{\top} A) \Pi_{\mathcal{F}} - \Pi_{\mathcal{F}} A^{\top} \\ A \Pi_{\mathcal{F}} & 0 \end{pmatrix},$$

viewed as a linear operator on $\mathcal{L}_{\mathcal{F}} \times \mathbb{R}^m$. If this operator is nonsingular, then there exist a neighborhood V_{red} of z^* in $\mathcal{L}_{\mathcal{F}} \times \mathbb{R}^m$ and a constant $\alpha_{\text{red}} > 0$ such that

$$\alpha_{\text{red}} \|z - z^*\| \leq \|\mathcal{T}_{\text{red}}(z)\|, \quad \forall z \in V_{\text{red}}.$$

In particular, after possibly shrinking V_{red} , $\mathcal{T}_{\text{red}}$ satisfies (23) with $Z_{\text{red}} = \{z^*\}$.

Proof. The formula (20) shows that $\mathcal{T}_{\text{red}}$ is continuously differentiable near z^* whenever f is C^2 near x^* . Differentiating the two components gives the displayed Jacobian.

Because $w^* \in W^*$ and Assumption 4 holds at x^* , the vector

$$\nabla f(x^*) + H_{\mathcal{F}} x^* + c_{\mathcal{F}} - A^{\top} y^*$$

belongs to $\mathcal{L}_{\mathcal{F}}^{\perp}$, and $Ax^* = b$. Thus $\mathcal{T}_{\text{red}}(z^*) = 0$.

If $D\mathcal{T}_{\text{red}}(z^*)$ is nonsingular, then by the inverse function theorem $\mathcal{T}_{\text{red}}$ is locally invertible around z^* , and its local inverse is Lipschitz. Since $\mathcal{T}_{\text{red}}(z^*) = 0$, there exist V_{red} and $c_{\text{red}} > 0$ such that

$$\|z - z^*\| \leq c_{\text{red}} \|\mathcal{T}_{\text{red}}(z) - \mathcal{T}_{\text{red}}(z^*)\| = c_{\text{red}} \|\mathcal{T}_{\text{red}}(z)\|, \quad \forall z \in V_{\text{red}}.$$

Taking $\alpha_{\text{red}} = 1/c_{\text{red}}$ gives the claimed local error bound. $\square$

Combining finite identification with the reduced residual identity gives the following consequence for the identified shadow points.

Corollary 3 *Let $F \in \{\widehat{\mathcal{T}}, \widehat{\mathcal{T}}_{\text{lin}}\}$, and let $\widehat{w}^j = F(w^j)$ be a shadow sequence satisfying the assumptions of Theorem 4. Suppose that Assumption 4 holds and that the identified manifold locally coincides with the affine set $\mathcal{F}$. Then there exists $J_{\text{red}} \geq 0$ such that, for all $j \geq J_{\text{red}}$,*

$$\widehat{x}^j \in \mathcal{F}, \quad \widehat{z}^j := (\widehat{x}^j - x^*, \widehat{y}^j) \in \mathcal{L}_{\mathcal{F}} \times \mathbb{R}^m,$$

and

$$\text{dist}(0, \mathcal{T}(\widehat{w}^j)) = \|\mathcal{T}_{\text{red}}(\widehat{z}^j)\|, \quad \forall j \geq J_{\text{red}}. \quad (24)$$

Proof. Finite identification gives $\widehat{x}^j \in \mathcal{F}$ for all sufficiently large j . Since $\widehat{w}^j \rightarrow w^*$, increasing J_{red} if necessary also gives $\widehat{z}^j \in U_{\text{red}}$ for all $j \geq J_{\text{red}}$. Therefore $\widehat{w}^j = \Psi(\widehat{z}^j)$ for all large j , and (24) follows from Theorem 5. $\square$

Remark 10 (Residual control by fixed-point steps) The same inclusions give a direct residual control by the fixed-point step. In the exact dPPM case $F = \widehat{\mathcal{T}}$,

$$\widehat{w}^j = (\mathcal{M} + \mathcal{T})^{-1} \mathcal{M} w^j$$

implies

$$\mathcal{M}(w^j - \widehat{w}^j) \in \mathcal{T}(\widehat{w}^j).$$

Therefore

$$\|\mathcal{T}_{\text{red}}(\widehat{z}^j)\| = \text{dist}(0, \mathcal{T}(\widehat{w}^j)) \leq \|\mathcal{M}(w^j - \widehat{w}^j)\| \leq \|\mathcal{M}\| \|w^j - \widehat{w}^j\|, \quad j \geq J_{\text{red}}.$$

In the linearized PFBS case $F = \widehat{\mathcal{T}}_{\text{lin}}$,

$$\widehat{w}^j = (\mathcal{M} + \mathcal{A}_h)^{-1} (\mathcal{M} - \mathcal{B}_h) w^j$$

implies

$$\mathcal{M}(w^j - \widehat{w}^j) + \mathcal{B}_h(\widehat{w}^j) - \mathcal{B}_h(w^j) \in (\mathcal{A}_h + \mathcal{B}_h)(\widehat{w}^j) = \mathcal{T}(\widehat{w}^j).$$

Therefore

$$\|\mathcal{T}_{\text{red}}(\widehat{z}^j)\| = \text{dist}(0, \mathcal{T}(\widehat{w}^j)) \leq \|\mathcal{M}(w^j - \widehat{w}^j) + \mathcal{B}_h(\widehat{w}^j) - \mathcal{B}_h(w^j)\|.$$

Since $\mathcal{B}_h(x, y) = (\nabla h(x), 0)$ and ∇h is L_h -Lipschitz,

$$\|\mathcal{B}_h(\widehat{w}^j) - \mathcal{B}_h(w^j)\| \leq L_h \|\widehat{x}^j - x^j\| \leq L_h \|\widehat{w}^j - w^j\|.$$

Consequently, for all $j \geq J_{\text{red}}$,

$$\|\mathcal{T}_{\text{red}}(\widehat{z}^j)\| \leq (\|\mathcal{M}\| + L_h) \|w^j - \widehat{w}^j\|.$$

4.3 Perturbed Reduction of the Halpern State

The previous subsection gives exact reduced residual identities at identified shadow points. The Halpern state itself is different: it may retain a normal component relative to the identified affine face. The next proposition records this distinction in a compact form. It is used only as a local structural description; the restart analysis in Section 5 is formulated in terms of fixed-point residuals.

Assume throughout this subsection that Assumption 4 holds. Let $F \in \{\widehat{\mathcal{T}}, \widehat{\mathcal{T}}_{\text{lin}}\}$ and let $\widehat{w}^j = F(w^j)$, with $w^j \rightarrow w^*$ and $\widehat{w}^j \rightarrow w^*$. Suppose that the shadow sequence has already identified $\mathcal{F}$ and that $\widehat{x}^j \in \mathcal{F}$ for all sufficiently large j . For such j , decompose the Halpern state as

$$\begin{aligned} x^j &= x^* + u^j + n^j, \\ u^j &:= \Pi_{\mathcal{F}}(x^j - x^*) \in \mathcal{L}_{\mathcal{F}}, \\ n^j &:= (I - \Pi_{\mathcal{F}})(x^j - x^*) \in \mathcal{L}_{\mathcal{F}}^{\perp}. \end{aligned}$$

Since $\widehat{x}^j \in \mathcal{F}$, write $\widehat{x}^j = x^* + \widehat{u}^j$, with $\widehat{u}^j \in \mathcal{L}_{\mathcal{F}}$. Define $z^j := (u^j, y^j)$ and $\widehat{z}^j := (\widehat{u}^j, \widehat{y}^j)$, and introduce the reduced metric operator

$$M_{\text{red}} := \begin{pmatrix} \Pi_{\mathcal{F}} P \Pi_{\mathcal{F}} & \Pi_{\mathcal{F}} A^{\top} \\ A \Pi_{\mathcal{F}} & Q \end{pmatrix}.$$

Finally, define the normal defect operator

$$D_{\perp} n := \begin{pmatrix} \Pi_{\mathcal{F}} P n \\ A n \end{pmatrix}, \quad n \in \mathcal{L}_{\mathcal{F}}^{\perp}.$$

For the linearized PFBS map, also define the reduced forward operator

$$\mathcal{B}_{\text{red},h}(u, y) := \begin{pmatrix} \Pi_{\mathcal{F}} \nabla h(x^* + u) \\ 0 \end{pmatrix},$$

and the reduced backward operator

$$\mathcal{A}_{\text{red},h}(u, y) := \begin{pmatrix} \Pi_{\mathcal{F}} \left(\nabla f(x^* + u) - \nabla h(x^* + u) + H_{\mathcal{F}}(x^* + u) \right) \\ + c_{\mathcal{F}} - A^{\top} y + \sigma A^{\top} A u \\ A u \end{pmatrix}.$$

Then $\mathcal{T}_{\text{red}} = \mathcal{A}_{\text{red},h} + \mathcal{B}_{\text{red},h}$.

Proposition 12 (Perturbed reduced state relations) *For every sufficiently large j , the following statements hold.*

(i) *If $F = \widehat{\mathcal{T}}$, then*

$$M_{\text{red}} z^j = M_{\text{red}} \widehat{z}^j + \mathcal{T}_{\text{red}}(\widehat{z}^j) - D_{\perp} n^j. \quad (25)$$

(ii) If $F = \widehat{\mathcal{T}}_{\text{lin}}$, then

$$M_{\text{red}} z^j - \mathcal{B}_{\text{red},h}(z^j) = M_{\text{red}} \widehat{z}^j + \mathcal{A}_{\text{red},h}(\widehat{z}^j) - D_{\perp} n^j + E_h^j, \quad (26)$$

where

$$E_h^j := \begin{pmatrix} \Pi_{\mathcal{F}}(\nabla h(x^* + u^j + n^j) - \nabla h(x^* + u^j)) \\ 0 \end{pmatrix}.$$

If ∇h is locally Lipschitz near x^* , then

$$\|E_h^j\| = O(\|n^j\|).$$

In both cases, when $n^j = 0$ the normal perturbation vanishes and the projected state relation becomes an autonomous reduced relation.

Proof. Write $\Pi = \Pi_{\mathcal{F}}$ for readability. We consider only indices for which $\widehat{x}^j \in \mathcal{F}$.

For the exact map, the resolvent relation

$$\mathcal{M}w^j \in \mathcal{M}\widehat{w}^j + \mathcal{T}(\widehat{w}^j)$$

gives the primal and dual shadow optimality conditions

$$\begin{aligned} 0 &\in \nabla f(\widehat{x}^j) + \partial g(\widehat{x}^j) - A^{\top} y^j + \sigma A^{\top} (A\widehat{x}^j - b) + P(\widehat{x}^j - x^j), \\ Q(\widehat{y}^j - y^j) &= b - A(2\widehat{x}^j - x^j). \end{aligned}$$

Using the affine-face representation, substituting $\widehat{x}^j = x^* + \widehat{u}^j$ and $x^j = x^* + u^j + n^j$, and projecting the primal condition onto $\mathcal{L}_{\mathcal{F}}$ gives

$$\begin{aligned} \Pi P u^j + \Pi A^{\top} y^j &= \Pi P \widehat{u}^j + \Pi \left(\nabla f(\widehat{x}^j) + H_{\mathcal{F}} \widehat{x}^j + c_{\mathcal{F}} + \sigma A^{\top} A \widehat{u}^j \right) - \Pi P n^j, \\ A u^j + Q y^j &= 2A \widehat{u}^j + Q \widehat{y}^j - A n^j. \end{aligned}$$

By the definitions of M_{red} , $\mathcal{T}_{\text{red}}$, and $D_{\perp}$, these two identities are exactly (25).

For the linearized map, the PFBS relation

$$(\mathcal{M} - \mathcal{B}_h)w^j \in (\mathcal{M} + \mathcal{A}_h)\widehat{w}^j$$

gives the same dual equation and the primal condition

$$0 \in \partial g(\widehat{x}^j) + \nabla f(\widehat{x}^j) - \nabla h(\widehat{x}^j) + \nabla h(x^j) - A^{\top} y^j + \sigma A^{\top} (A\widehat{x}^j - b) + P(\widehat{x}^j - x^j).$$

Projecting this condition onto $\mathcal{L}_{\mathcal{F}}$, using the same dual equation as above, and adding and subtracting $\Pi \nabla h(x^* + u^j)$ gives

$$\begin{aligned} \Pi P u^j + \Pi A^{\top} y^j - \Pi \nabla h(x^* + u^j) &= \Pi P \widehat{u}^j + \Pi (\nabla f(\widehat{x}^j) - \nabla h(\widehat{x}^j)) \\ &\quad + \Pi (H_{\mathcal{F}} \widehat{x}^j + c_{\mathcal{F}} + \sigma A^{\top} A \widehat{u}^j) - \Pi P n^j \\ &\quad + \Pi (\nabla h(x^* + u^j + n^j) - \nabla h(x^* + u^j)), \\ A u^j + Q y^j &= 2A \widehat{u}^j + Q \widehat{y}^j - A n^j. \end{aligned}$$

This is precisely (26). Since $w^j \rightarrow w^*$, both $x^* + u^j + n^j$ and $x^* + u^j$ lie in any fixed neighborhood of x^* for all sufficiently large j . The local Lipschitz continuity of ∇h therefore gives

$$\|E_h^j\| \leq L\|n^j\|$$

for some local Lipschitz constant L , proving $\|E_h^j\| = O(\|n^j\|)$. $\square$

Remark 11 (Normal component induced by reflection and anchoring) The perturbation in Proposition 12 is caused by the fact that the Halpern state x^j need not lie on the identified affine face, even after the shadow point $\hat{x}^j$ has identified it.

To see this explicitly, consider one epoch of a restarted reflected Halpern scheme. Let

$$\nu_{n,k} := (I - \Pi_{\mathcal{F}})(x_{n,k} - x^*)$$

be the normal component of the Halpern state, and suppose that $\hat{x}_{n,k} \in \mathcal{F}$. Then

$$(I - \Pi_{\mathcal{F}})(\hat{x}_{n,k} - x^*) = 0.$$

The reflected point satisfies

$$\bar{x}_{n,k} = (1 + \gamma)\hat{x}_{n,k} - \gamma x_{n,k},$$

and therefore

$$\bar{\nu}_{n,k} := (I - \Pi_{\mathcal{F}})(\bar{x}_{n,k} - x^*) = -\gamma\nu_{n,k}.$$

The Halpern update

$$x_{n,k+1} = \frac{1}{k+2}x_{n,0} + \frac{k+1}{k+2}\bar{x}_{n,k}$$

gives

$$\nu_{n,k+1} = \frac{1}{k+2}\nu_{n,0} - \frac{\gamma(k+1)}{k+2}\nu_{n,k}.$$

This linear recurrence also gives the decay rate of the normal drift within such an identified epoch. Setting $s_{n,k} := (k+1)\nu_{n,k}$, we obtain

$$s_{n,k+1} = \nu_{n,0} - \gamma s_{n,k}.$$

Solving this recursion gives, for every fixed $\gamma \in (-1, 1]$,

$$s_{n,k} = \frac{1 + \gamma(-\gamma)^k}{1 + \gamma}\nu_{n,0}, \quad \nu_{n,k} = \frac{1 + \gamma(-\gamma)^k}{(1 + \gamma)(k+1)}\nu_{n,0}.$$

Thus, throughout the reflected range $\gamma \in (-1, 1]$, the normal component decays as $O(1/k)$ along the epoch, although it need not vanish in finite time. An identically zero normal sequence is obtained only under the invariance condition that the epoch anchor lies on the affine face. Consequently, the autonomous reduced dynamics without the perturbation term are valid for the Halpern state only under the additional invariance condition $n^j = 0$. Without this condition, the correct late-stage description is the perturbed reduced representation (25) or (26).

Remark 12 (Scope of the affine-face model) Assumption 4 is stronger than partial smoothness or piecewise linear-quadratic structure alone. It is appropriate when, on the identified face, g is locally a quadratic function plus the indicator of an affine set, or more generally when the identified normal fiber coincides with all of $\mathcal{L}_{\mathcal{F}}^\perp$. For generic ℓ_1 regularizers, nonnegativity constraints, box constraints, and simplex constraints, the fixed normal fiber after identification is usually a cone or a strict subset of $\mathcal{L}_{\mathcal{F}}^\perp$ rather than the whole normal space. In those cases the tangential projected dynamics remain informative, but the exact residual identity (21) should be replaced by a polyhedral error-bound or metric-subregularity argument.

The results in this section clarify the role of identification in reflected Halpern and restarted reflected Halpern schemes. The finite identification property belongs to the shadow sequence $\hat{x}^j = F(w^j)$, not to the Halpern state x^j itself. After identification, the KKT residual at the shadow points is exactly represented by the reduced residual $\|\mathcal{T}_{\text{red}}(\hat{z}^j)\|$ on the tangent space of the active affine face. This provides the local geometric mechanism needed for the restart analysis in Section 5.

At the same time, the Halpern state may retain a nonzero normal component. The resulting late-stage dynamics are therefore reduced only up to a normal perturbation, unless the algorithm is initialized or restarted directly on the identified face. This distinction is essential for a correct local theory of reflected Halpern primal–dual methods.

5 Restart Strategies and Linear Convergence under Sharpness

Restart converts sublinear residual decay into a linear epoch-wise contraction when the fixed-point residual satisfies a sharpness condition on the set visited by the restarted trajectory. We develop this conditional analysis for a generic map

$$F \in \{\hat{\mathcal{T}}, \hat{\mathcal{T}}_{\text{lin}}\},$$

covering both exact and linearized augmented primal–dual schemes. We prove fixed-frequency and adaptive restart guarantees, and then verify the required sharpness from KKT error bounds, including the reduced error-bound mechanism on the identified face. Thus the restart theorems are global only when sharpness holds on the full relevant trajectory set. In the applications below, sharpness is obtained from local error bounds, and the conclusions are local or tail linear convergence statements after the trajectory has entered the corresponding neighborhood.

5.1 Restarted Reflected Halpern Under Sharpness

Fix one of these two maps, together with its associated preconditioner $\mathcal{M}$, and let

$$\mathcal{H} := \mathbb{R}^n \times \mathbb{R}^m, \quad F \in \{\hat{\mathcal{T}}, \hat{\mathcal{T}}_{\text{lin}}\}.$$

For both choices of F , the fixed-point set coincides with the KKT set $W^\star = \mathcal{T}^{-1}(0)$ introduced in Section 4. Throughout this section, the relaxation parameter γ is chosen from the admissible range for the selected map; the reflected map

$$\mathcal{S}_\gamma := (1 + \gamma)F - \gamma I$$

is $\mathcal{M}$ -nonexpansive. Concretely, this means $\gamma \in (-1, 1]$ for the exact dPPM map and $\gamma \in (-1, 1 - L_h/(2\mu_x)]$ for the linearized PFBS map.

For $w \in \mathcal{H}$, define the fixed-point residual

$$r_F(w) := \|w - F(w)\|_{\mathcal{M}}.$$

Since

$$w - \mathcal{S}_\gamma(w) = (1 + \gamma)(w - F(w)),$$

one has

$$\|w - \mathcal{S}_\gamma(w)\|_{\mathcal{M}} = (1 + \gamma)r_F(w). \quad (27)$$

Thus the residual of the reflected map and the residual of F are equivalent up to the scalar $1 + \gamma$.

Definition 2 (Sharpness) Let $\mathcal{K} \subset \mathcal{H}$. We say that F satisfies the α -sharpness condition on $\mathcal{K}$ if $\alpha > 0$ and

$$\alpha \operatorname{dist}_{\mathcal{M}}(w, W^\star) \leq r_F(w), \quad \forall w \in \mathcal{K}, \quad (28)$$

where

$$\operatorname{dist}_{\mathcal{M}}(w, W^\star) := \inf_{w^\star \in W^\star} \|w - w^\star\|_{\mathcal{M}}.$$

When $\mathcal{M}$ is singular, $\operatorname{dist}_{\mathcal{M}}$ is understood as the semidistance induced by the $\mathcal{M}$ -seminorm; positive definiteness is imposed explicitly whenever a genuine distance is required.

The restarted reflected Halpern scheme is organized into epochs. Given the anchor $w_{n,0}$ of epoch n , the inner iterates are

$$w_{n,k+1} = \frac{1}{k+2}w_{n,0} + \frac{k+1}{k+2}\mathcal{S}_\gamma(w_{n,k}), \quad k \geq 0. \quad (29)$$

When an epoch stops at length K_n , the next anchor is

$$w_{n+1,0} := w_{n,K_n}. \quad (30)$$

Algorithm 3 records only the outer restart wrapper. It is used with either the exact dPPM map of Algorithm 1 or the linearized PFBS map of Algorithm 2; the inner reflected Halpern step is the epoch iteration (29).

We use two restart rules. The first is a fixed-frequency rule. If a sharpness constant α is known and a target contraction factor $\beta \in (0, 1)$ is chosen, set

$$K_n \equiv k^\star, \quad k^\star := \left\lceil \frac{2}{\alpha\beta(1+\gamma)} \right\rceil. \quad (31)$$

Algorithm 3: Restart Wrapper for Reflected Halpern Schemes

```

1: Input: map  $F$ , relaxation  $\gamma$ , initial anchor  $w_{0,0}$ , restart rule.
2: for  $n = 0, 1, 2, \dots$  do
3:   Run (29) from anchor  $w_{n,0}$  until the restart rule is met.
4:   Denote the stopping index by  $K_n$ .
5:   Set  $w_{n+1,0} \leftarrow w_{n,K_n}$ .
6: end for

```

The second is an adaptive rule based only on the observed residual. Given $\beta \in (0, 1)$ and an initial epoch length $K_0 \in \mathbb{N}$, restart when

$$\begin{cases} r_F(w_{n,k}) \leq \beta r_F(w_{n,0}), & n \geq 1, \\ k \geq K_0, & n = 0. \end{cases} \quad (32)$$

The single estimate needed for both restart rules is the following epochwise residual bound.

Lemma 1 (Epoch residual bound) *For every epoch $n \geq 0$ and every $k \geq 0$,*

$$(1 + \gamma)r_F(w_{n,k}) = \|w_{n,k} - \mathcal{S}_\gamma(w_{n,k})\|_{\mathcal{M}} \leq \frac{2}{k+1} \text{dist}_{\mathcal{M}}(w_{n,0}, W^*).$$

Equivalently,

$$r_F(w_{n,k}) \leq \frac{2}{(1 + \gamma)(k+1)} \text{dist}_{\mathcal{M}}(w_{n,0}, W^*). \quad (33)$$

Proof. For the admissible choices of γ , the map $\mathcal{S}_\gamma$ is $\mathcal{M}$ -nonexpansive and

$$\text{Fix}(\mathcal{S}_\gamma) = \text{Fix}(F) = W^*.$$

The residual estimates in Propositions 4 and 8 apply to the two possible choices of F . Applied to the inner iteration (29), they give, for every $w^* \in W^*$,

$$\|w_{n,k} - \mathcal{S}_\gamma(w_{n,k})\|_{\mathcal{M}} \leq \frac{2}{k+1} \|w_{n,0} - w^*\|_{\mathcal{M}}.$$

Taking the infimum over $w^* \in W^*$ and using (27) proves the claim. $\square$

5.2 Linear Convergence of Restarted Reflected Halpern Schemes

We now show that sharpness converts the $O(1/k)$ inner residual decay into a linear contraction across restart epochs.

Theorem 6 (Fixed-frequency restart) *Let the restart points $\{w_{n,0}\}_{n \geq 0}$ be generated by (29) and (30) with the fixed epoch length (31) for a prescribed $\beta \in (0, 1)$. Assume that all iterates remain in a set $\mathcal{K} \subset \mathcal{H}$ on which F satisfies the α -sharpness condition (28). Then*

$$\text{dist}_{\mathcal{M}}(w_{n,0}, W^*) \leq \beta^n \text{dist}_{\mathcal{M}}(w_{0,0}, W^*), \quad \forall n \geq 0. \quad (34)$$

Proof. By (30), $w_{n+1,0} = w_{n,k^*}$. Since the iterates lie in $\mathcal{K}$, sharpness at w_{n,k^*} gives

$$\text{dist}_{\mathcal{M}}(w_{n+1,0}, W^*) \leq \frac{1}{\alpha} r_F(w_{n,k^*}).$$

Using (33),

$$\text{dist}_{\mathcal{M}}(w_{n+1,0}, W^*) \leq \frac{2}{\alpha(1+\gamma)(k^*+1)} \text{dist}_{\mathcal{M}}(w_{n,0}, W^*).$$

The choice of k^* in (31) implies

$$\frac{2}{\alpha(1+\gamma)(k^*+1)} \leq \frac{2}{\alpha(1+\gamma)k^*} \leq \beta.$$

Therefore the restart anchors contract by a factor not exceeding β at every epoch, and iteration gives (34). $\square$

The adaptive rule does not require prior knowledge of α . For this result we assume $\mathcal{M} \succ 0$; then $\text{dist}_{\mathcal{M}}$ is a genuine distance and the residual-to-distance ratio below is well defined away from the solution set.

Theorem 7 (Adaptive restart) *Let the restart points $\{w_{n,0}\}_{n \geq 0}$ be generated by (29) and (30) with the adaptive rule (32). Assume that $\mathcal{M} \succ 0$, that all iterates remain in a set $\mathcal{K} \subset \mathcal{H}$, and that F satisfies the α -sharpness condition (28) on $\mathcal{K}$. Then the following statements hold.*

(i) *Every epoch $n \geq 1$ terminates after finitely many inner iterations. If $w_{n,0} \notin W^*$, define*

$$\alpha_n := \frac{r_F(w_{n,0})}{\text{dist}_{\mathcal{M}}(w_{n,0}, W^*)},$$

then $\alpha_n \geq \alpha$ and the stopping index satisfies

$$K_n \leq \left\lceil \frac{2}{\alpha_n \beta (1+\gamma)} \right\rceil \leq \left\lceil \frac{2}{\alpha \beta (1+\gamma)} \right\rceil. \quad (35)$$

If $w_{n,0} \in W^$, then $r_F(w_{n,0}) = 0$ and the restart condition holds at $k = 0$.*

(ii) *The fixed-point residuals at the restart points decay linearly:*

$$r_F(w_{n,0}) \leq \beta^{n-1} r_F(w_{1,0}), \quad \forall n \geq 1. \quad (36)$$

(iii) *The restart points converge linearly in $\mathcal{M}$ -distance:*

$$\text{dist}_{\mathcal{M}}(w_{n,0}, W^*) \leq \frac{2}{\alpha(1+\gamma)(K_0+1)} \beta^{n-1} \text{dist}_{\mathcal{M}}(w_{0,0}, W^*), \quad \forall n \geq 1. \quad (37)$$

Proof. We prove the three claims in order. For (i), fix $n \geq 1$. If $w_{n,0} \in W^\star$, then $r_F(w_{n,0}) = 0$, and the adaptive rule is satisfied at $k = 0$. Otherwise, sharpness gives $\alpha_n \geq \alpha$. By (33), for every $k \geq 0$,

$$r_F(w_{n,k}) \leq \frac{2}{(1+\gamma)(k+1)} \operatorname{dist}_{\mathcal{M}}(w_{n,0}, W^\star) = \frac{2}{\alpha_n(1+\gamma)(k+1)} r_F(w_{n,0}).$$

Thus the condition $r_F(w_{n,k}) \leq \beta r_F(w_{n,0})$ holds once $k+1 \geq 2/(\alpha_n \beta (1+\gamma))$, proving finite termination and (35).

For (ii), the adaptive rule and the restart update imply, for every $n \geq 1$,

$$r_F(w_{n+1,0}) = r_F(w_{n,K_n}) \leq \beta r_F(w_{n,0}).$$

Iterating this inequality proves (36).

For (iii), sharpness at $w_{n,0}$ and (36) give

$$\operatorname{dist}_{\mathcal{M}}(w_{n,0}, W^\star) \leq \frac{1}{\alpha} \beta^{n-1} r_F(w_{1,0}), \quad n \geq 1.$$

Since the initial epoch has length K_0 , $w_{1,0} = w_{0,K_0}$, and (33) yields

$$r_F(w_{1,0}) \leq \frac{2}{(1+\gamma)(K_0+1)} \operatorname{dist}_{\mathcal{M}}(w_{0,0}, W^\star).$$

Combining the last two inequalities proves (37). $\square$

5.3 Verifying Sharpness from KKT Error Bounds

The restart theorems above require the fixed-point sharpness condition (28). We now connect this condition with standard KKT error bounds. This is the only verification direction needed for the restart theory. In particular, we do not require a converse equivalence between fixed-point sharpness and metric subregularity.

Let $U \subset \mathcal{H}$. We say that the augmented KKT mapping satisfies an error bound on U if there exists $\alpha_U > 0$ such that

$$\alpha_U \operatorname{dist}(u, W^\star) \leq \operatorname{dist}(0, \mathcal{T}(u)), \quad \forall u \in U. \quad (38)$$

This is the usual metric-subregularity estimate, restricted to the set U .

The first lemma converts Euclidean sharpness estimates into the $\mathcal{M}$ -metric sharpness used in the restart analysis.

Lemma 2 (Metric conversion of sharpness) *Assume $\mathcal{M} \succ 0$, and let $\lambda_{\min}(\mathcal{M})$ and $\lambda_{\max}(\mathcal{M})$ denote its extremal eigenvalues. If a mapping F satisfies*

$$\alpha_E \operatorname{dist}(w, W^\star) \leq \|w - F(w)\|, \quad \forall w \in \mathcal{K},$$

then F satisfies (28) on $\mathcal{K}$ with

$$\alpha_M := \alpha_E \sqrt{\frac{\lambda_{\min}(\mathcal{M})}{\lambda_{\max}(\mathcal{M})}}.$$

Proof. For every w ,

$$\text{dist}_{\mathcal{M}}(w, W^*) \leq \sqrt{\lambda_{\max}(\mathcal{M})} \text{dist}(w, W^*),$$

whereas

$$r_F(w) = \|w - F(w)\|_{\mathcal{M}} \geq \sqrt{\lambda_{\min}(\mathcal{M})} \|w - F(w)\|.$$

Combining these inequalities with the Euclidean sharpness estimate proves the claim. $\square$

Proposition 13 (KKT error bounds imply fixed-point sharpness) *Assume that the KKT error bound (38) holds on a set $U \subset \mathcal{H}$. Let $\mathcal{K} \subset \mathcal{H}$ satisfy*

$$\widehat{\mathcal{T}}(\mathcal{K}) \cup \widehat{\mathcal{T}}_{\text{lin}}(\mathcal{K}) \subset U.$$

Then the following statements hold.

(i) *The exact dPPM map satisfies*

$$\alpha_{\text{ex},U}^E \text{dist}(w, W^*) \leq \|w - \widehat{\mathcal{T}}(w)\|, \quad \forall w \in \mathcal{K},$$

with

$$\alpha_{\text{ex},U}^E := \frac{\alpha_U}{\alpha_U + \|\mathcal{M}\|}.$$

(ii) *The linearized PFBS map satisfies*

$$\alpha_{\text{lin},U}^E \text{dist}(w, W^*) \leq \|w - \widehat{\mathcal{T}}_{\text{lin}}(w)\|, \quad \forall w \in \mathcal{K},$$

with

$$\alpha_{\text{lin},U}^E := \frac{\alpha_U}{\alpha_U + \|\mathcal{M}\| + L_h}.$$

(iii) *If $\mathcal{M} \succ 0$, then both maps satisfy the sharpness condition (28) on $\mathcal{K}$, with the constants from parts (i)–(ii) multiplied by $\sqrt{\lambda_{\min}(\mathcal{M})/\lambda_{\max}(\mathcal{M})}$.*

Proof. For the exact map, fix $w \in \mathcal{K}$ and set $u := \widehat{\mathcal{T}}(w)$. Then $u \in U$, and the resolvent relation gives

$$\mathcal{M}(w - u) \in \mathcal{T}(u).$$

Therefore

$$\alpha_U \text{dist}(u, W^*) \leq \text{dist}(0, \mathcal{T}(u)) \leq \|\mathcal{M}(w - u)\| \leq \|\mathcal{M}\| \|w - u\|.$$

The triangle inequality yields

$$\text{dist}(w, W^*) \leq \|w - u\| + \text{dist}(u, W^*) \leq \left(1 + \frac{\|\mathcal{M}\|}{\alpha_U}\right) \|w - u\|,$$

which proves part (i).

For the linearized map, set $u := \widehat{\mathcal{T}}_{\text{lin}}(w)$. The PFBS relation gives

$$(\mathcal{M} - \mathcal{B}_h)(w) \in (\mathcal{M} + \mathcal{A}_h)(u),$$

and therefore

$$\mathcal{M}(w - u) + \mathcal{B}_h(u) - \mathcal{B}_h(w) \in (\mathcal{A}_h + \mathcal{B}_h)(u) = \mathcal{T}(u).$$

Since $u \in U$, the error bound and the L_h -Lipschitz continuity of $\mathcal{B}_h$ give

$$\begin{aligned} \alpha_U \operatorname{dist}(u, W^*) &\leq \operatorname{dist}(0, \mathcal{T}(u)) \\ &\leq \|\mathcal{M}(w - u) + \mathcal{B}_h(u) - \mathcal{B}_h(w)\| \\ &\leq (\|\mathcal{M}\| + L_h)\|w - u\|. \end{aligned}$$

The same triangle-inequality argument proves part (ii). Part (iii) follows from Lemma 2. $\square$

Remark 13 (Local-to-tail interpretation) Proposition 13 is local in the same sense as metric subregularity. The error-bound set U need only contain the forward images of the part of the trajectory on which the restart theorem is applied. Thus, if a KKT error bound holds in a neighborhood U of a solution w^* , and if a relevant tail set $\mathcal{K}_{\text{tail}}$ of restart points satisfies

$$\widehat{\mathcal{T}}(\mathcal{K}_{\text{tail}}) \cup \widehat{\mathcal{T}}_{\text{lin}}(\mathcal{K}_{\text{tail}}) \subset U,$$

then the fixed-point sharpness condition holds on this tail set in the positive definite metric setting. Since both maps are continuous under the standing assumptions of Sections 2–3, this inclusion is automatic for all sufficiently late points in such a tail set whenever the restarted trajectory converges to w^* . The restart results should therefore be read as local linear convergence statements after the trajectory has entered the error-bound neighborhood.

Proposition 14 (Box-constrained linear and quadratic programs) *Let*

$$C := [l, u] \subset \mathbb{R}^n, \quad g = \delta_C,$$

where C is a nonempty box. Suppose that either

$$f(x) = c^\top x \quad \text{or} \quad f(x) = \frac{1}{2}x^\top Hx + c^\top x, \quad H \succeq 0.$$

Then, for every $w^* \in W^*$, there exist a neighborhood U of w^* and a constant $\alpha_U > 0$ such that the augmented KKT mapping satisfies

$$\alpha_U \operatorname{dist}(w, W^*) \leq \operatorname{dist}(0, \mathcal{T}(w)), \quad \forall w \in U.$$

Consequently, if $\mathcal{M} \succ 0$ and

$$\widehat{\mathcal{T}}(\mathcal{K}) \cup \widehat{\mathcal{T}}_{\text{lin}}(\mathcal{K}) \subset U,$$

then both the exact and linearized fixed-point maps satisfy the sharpness condition (28) on $\mathcal{K}$.

Proof. For the stated choices of f , write $\nabla f(x) = Hx + c$, with $H = 0$ in the linear case. Since $g = \delta_C$, we have $\partial g = N_C$, and

$$\mathcal{T}(x, y) = \begin{pmatrix} (H + \sigma A^\top A)x - A^\top y + c - \sigma A^\top b + N_C(x) \\ Ax - b \end{pmatrix}.$$

The graph of N_C is a finite union of polyhedral sets: each piece is obtained by fixing which lower bounds, upper bounds, and inactive box constraints are active. Adding the displayed linear terms and the equality component $Ax - b$ preserves piecewise polyhedrality. Thus $\mathcal{T}$ is a piecewise polyhedral multifunction.

The Hoffman–Robinson error-bound theorem for piecewise polyhedral multifunctions [20, 31] implies metric subregularity of $\mathcal{T}$ at $(w^*, 0)$. Thus there are a neighborhood U of w^* and a constant $\kappa_U > 0$ such that

$$\text{dist}(w, \mathcal{T}^{-1}(0)) \leq \kappa_U \text{dist}(0, \mathcal{T}(w)), \quad \forall w \in U.$$

Since $W^* = \mathcal{T}^{-1}(0)$, the asserted error bound follows with $\alpha_U = 1/\kappa_U$. The final statement is exactly Proposition 13. $\square$

Corollary 4 (Sharpness after affine-face identification) *Suppose the hypotheses of Corollary 2 hold, and the augmented KKT mapping satisfies a face-restricted error bound on $V := \Psi(V_{\text{red}})$. Let $\mathcal{K}_{\text{late}} \subset \mathcal{H}$ be a set such that*

$$\widehat{\mathcal{T}}(\mathcal{K}_{\text{late}}) \cup \widehat{\mathcal{T}}_{\text{lin}}(\mathcal{K}_{\text{late}}) \subset V.$$

If $\mathcal{M} \succ 0$, then both fixed-point maps satisfy the sharpness condition (28) on $\mathcal{K}_{\text{late}}$.

Proof. Corollary 2 gives a KKT error bound of the form (38) on $U = V$. Applying Proposition 13 and the metric conversion in Lemma 2 proves the result. $\square$

Consequently, the restart theorems in Section 5.2 can be used in two ways. If sharpness is known directly for the fixed-point residual, Theorems 6 and 7 apply under their stated metric assumptions. If instead a KKT error bound or metric-subregularity estimate is available, Proposition 13 transfers it to fixed-point sharpness. Proposition 14 provides this verification for the box-constrained linear and convex quadratic programs used in the numerical section. In the identified affine-face regime, Corollary 4 supplies the same verification from the reduced residual theory of Section 4.

6 Numerical Experiments

This section reports computational evidence for the augmented primal–dual family developed in the paper. The experiments are organized by problem class and by algorithmic role. We do not benchmark every family member on every problem class. Instead, each comparison is chosen to ensure that the tested updates are computationally meaningful for the representation at hand: LP experiments focus on equality-form solvers of the PDHG and CP-AL types, while QP experiments test both subproblem-based and linearized variants.

6.1 Experimental Protocol and Method Scope

All reported GPU experiments were run on one H100 GPU with 80 GB HBM3 memory and CUDA 12.8.1. External baselines were run under the same hardware conditions whenever they appear in a direct comparison, with presolve disabled where stated. Within each benchmark family, all methods use the same stopping tolerance and wall-clock or iteration budget. We report solved counts, time-limit counts, total runtime when informative, and the shifted geometric mean

$$\text{SGM10}(t_1, \dots, t_N) := \exp\left(\frac{1}{N} \sum_{i=1}^N \log(t_i + 10)\right) - 10.$$

When a run reaches the time limit, the time limit is used in the computation of SGM10.

For the restarted variants proposed here, restart decisions are made using the fixed-point residual of the underlying map, as in Section 5. More precisely, during epoch n , let $w_{n,t}$ be the restart candidate at the t -th restart check and set

$$R_{n,t} := r_F(w_{n,t}), \quad r_F(w) := \|w - F(w)\|_{\mathcal{M}},$$

where F is the corresponding base map. By (27), this is equivalent, up to the scalar $1 + \gamma$, to using the residual of the reflected map. Following the adaptive restart criteria used in HPR-LP, cuPDLpx, and PDHCG-II [9, 24, 22], a restart is performed if one of the following conditions is met:

- (i) *Sufficient decay*: $R_{n,t} \leq \beta_{\text{suff}} R_{n,0}$.
- (ii) *Necessary decay plus no local progress*: $R_{n,t} \leq \beta_{\text{nec}} R_{n,0}$ and $R_{n,t} > R_{n,t-1}$.
- (iii) *Long inner loop*: $t \geq \beta_{\text{art}} k$, where k is the cumulative iteration count.

We use $(\beta_{\text{suff}}, \beta_{\text{nec}}, \beta_{\text{art}}) = (0.2, 0.8, 0.36)$ for the LP experiments and the corresponding PDHCG-II restart parameters for the QP experiments.

Remark 14 (Normalized CP-AL parametrization) For the LP experiments, we use the normalized one-parameter CP-AL subfamily

$$P = \sigma(\lambda I_n - A^\top A), \quad Q = \sigma^{-1} I_m,$$

or equivalently $\tau^{-1} = \sigma\lambda$ and $\rho = \sigma$. This is not a new algorithmic family, but a convenient parametrization in which λ controls the primal metric while σ acts simultaneously as augmentation parameter and dual stepsize. We choose λ slightly larger than an estimate of $\|A\|^2$, obtained by power iteration, and initialize $\sigma_0 = 0.5 \lambda^{-1/2}$. The parameter σ is updated only at restart epochs. If Δx and Δy are the primal and dual displacements over the completed epoch, define

$$d_x^2 := \lambda \|\Delta x\|^2 - \|A\Delta x\|^2, \quad \hat{\sigma} := \frac{\|\Delta y\|}{\sqrt{d_x^2}}.$$

When this estimate is numerically reliable, we use the damped log-scale update

$$\log \sigma^+ = (1 - \theta) \log \sigma + \theta \log \hat{\sigma}, \quad \theta = 0.5,$$

with a fixed safeguard interval proportional to $\lambda^{-1/2}$. If the estimate is unreliable, σ is left unchanged; the safeguard interval is used only to avoid degenerate scalings of the metric. Thus **RHR-CP-AL** denotes the restarted Halpern-reflected realization of the **CP-AL** base map equipped with this safeguarded adaptive σ rule.

For all experiments we report a relative KKT score associated with the model (1). Recall that its KKT conditions are

$$0 \in \nabla f(x) + \partial g(x) - A^\top y, \quad Ax = b.$$

All multipliers used in the residual evaluation are expressed in this sign convention. For a reported primal-dual pair (x, y) with finite objective value, define

$$r_p(x) := \max \left\{ \frac{\|Ax - b\|_\infty}{1 + \|b\|_\infty}, \frac{\text{dist}_\infty(x, \text{dom } g)}{1 + \|x\|_\infty} \right\},$$

$$r_d(x, y) := \frac{\text{dist}_\infty(A^\top y - \nabla f(x), \partial g(x))}{1 + \|\nabla f(x)\|_\infty + \|A^\top y\|_\infty}.$$

Here dist_∞ denotes distance in the ℓ_∞ norm, with the convention that the distance to the empty set is $+\infty$. The relative objective gap is

$$r_g(x, y) := \frac{|\Phi(x) - d(y)|}{1 + |\Phi(x)| + |d(y)|}, \quad d(y) := b^\top y + \inf_{z \in \mathbb{R}^n} \{f(z) + g(z) - \langle A^\top y, z \rangle\}.$$

We then set

$$r_{\text{KKT}} := \max\{r_p, r_d, r_g\}.$$

For the equality-box LP and QP test problems below, g is the indicator of the box $[l, u]$; these definitions reduce to the usual primal feasibility, dual feasibility, and relative primal-dual gap residuals. A run is declared solved when $r_p \leq \varepsilon_{\text{feas}}$, $r_d \leq \varepsilon_{\text{feas}}$, and $r_g \leq \varepsilon_{\text{opt}}$. When a table states a single tolerance ε , we use $\varepsilon_{\text{feas}} = \varepsilon_{\text{opt}} = \varepsilon$. When fixed-point residual traces are compared across methods, each residual is normalized by its initial value because the underlying fixed-point metrics differ.

We distinguish base maps from solver names. The names **PDHG**, **CP-AL**, **FA-CP**, **Lin-PDHG**, and **Lin-CP-AL** refer to operator maps or algorithmic families. In numerical tables, the prefix **RHR-** denotes the method obtained by applying the restarted reflected Halpern wrapper to the corresponding base map. External or previously published solvers keep their original names; in particular, the LP PDHG-type comparator is **cuPDLPx** [24], the LP Halpern baseline is **HPR-LP** [9], and the QP external solvers include **PDHCG-II** [22], **PDQP** [28], and **HPR-QP** [10]. This naming convention is used consistently below.

Table 2: Synthetic families for the equality-constrained box-LP test set.

Family	α_r	α_c	p_r	p_c	γ_{dep}	Interpretation
Baseline	2	2	0.02	0.02	0.98	Mild scaling and weak dependence
Ill-scaled	5	5	0.02	0.02	0.98	Severe scaling and weak dependence
Near-dependent	2	2	0.28	0.28	0.9995	Mild scaling and strong dependence
Hybrid	3	3	0.18	0.18	0.995	Simultaneous scaling and dependence

6.2 Linear Programming Experiments

We begin with equality-constrained box LPs of the form

$$\min_{x \in \mathbb{R}^n} c^\top x \quad \text{s.t.} \quad Ax = b, \quad l \leq x \leq u. \quad (39)$$

This is the most direct LP realization of the framework: the equality constraint is represented natively, and the augmentation acts directly on the residual $Ax - b$. The LP subsection has three roles. First, synthetic families isolate scaling and near dependence in A . Second, a strict MIPLIB subset provides a standard benchmark comparison. Third, an ablation trace illustrates the two-stage behavior created by restart, Halpern anchoring, and reflection.

The synthetic instances are generated by sampling a sparse matrix $A \in \mathbb{R}^{m \times n}$, enforcing nonzero rows and columns, injecting near dependence among selected rows and columns, and then applying diagonal row and column scalings. A feasible point $x^{\text{fea}} \in [l, u]$ is planted and $b := Ax^{\text{fea}}$. Within each family we generate 1000 instances, consisting of 400 small, 400 medium, and 200 large problems with

$$(m, n, \text{density}) \in \{(150, 400, 0.045), (250, 700, 0.025), (400, 1000, 0.012)\}.$$

The four synthetic LP families are defined in Table 2. Here α_r and α_c are the logarithmic row and column scaling exponents, p_r and p_c are the fractions of rows and columns used in the dependence step, and γ_{dep} is the dependence coefficient.

For this synthetic LP class, the main comparison is between **cuPDLPx** and **RHR-CP-AL**. The terminal score is $s := r_{\text{KKT}}$. All methods are run without presolve, with $\varepsilon_{\text{opt}} = \varepsilon_{\text{feas}} = 10^{-8}$, iteration limit 10^5 , and time limit 60 seconds. Table 3 reports the median terminal score and the number of instances satisfying two accuracy thresholds. The table is designed to isolate the effect of equality geometry; the emphasis is on terminal quality rather than wall-clock time. The advantage of **RHR-CP-AL** is strongest in the near-dependent and hybrid families, exactly where the equality geometry is most degenerate.

We next turn to real benchmark instances. Starting from MIPLIB 2017 [17], we extract the 14 problems whose original LP formulation already matches (39). All runs are performed without presolve; the methods are therefore applied to the same native equality-form representation. In addition to **cuPDLPx**, we include **HPR-LP** [9] as a Halpern-restarted LP baseline; it is called through its direct **QPSReader** MPS interface, without invoking **HiGHS** or LP presolve.

Table 3: Equality-constrained box-LP results on 1000 instances per family. For each family, we report the median terminal score s and the numbers of instances satisfying $s \leq 10^{-4}$ and $s \leq 10^{-6}$.

Family	Median s		$\#\{s \leq 10^{-4}\}$		$\#\{s \leq 10^{-6}\}$	
	cuPDLPx	RHR-CP-AL	cuPDLPx	RHR-CP-AL	cuPDLPx	RHR-CP-AL
Baseline	9.42×10^{-9}	9.69×10^{-9}	999	1000	997	996
Ill-scaled	9.75×10^{-9}	6.78×10^{-8}	961	991	871	933
Near-dependent	3.37×10^{-2}	5.68×10^{-6}	163	926	1	64
Hybrid	1.83×10^{-6}	8.02×10^{-7}	748	993	456	557

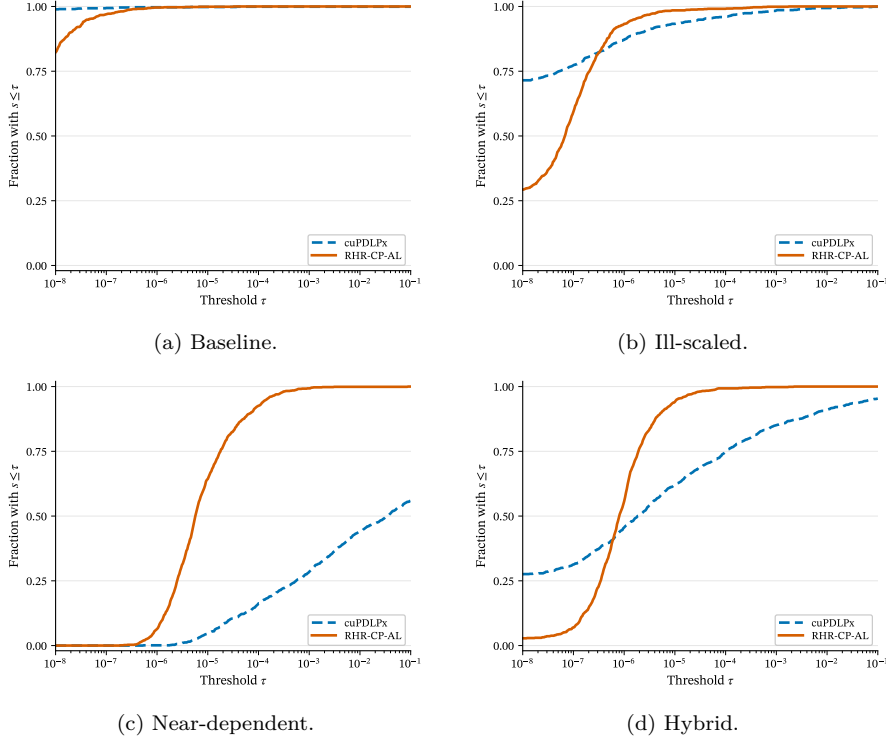


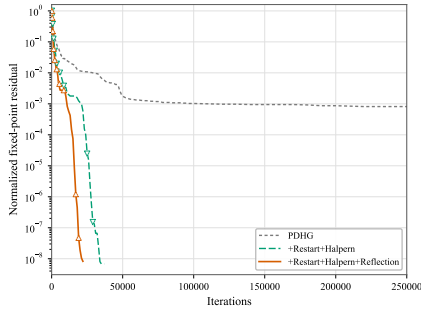
Fig. 1: Empirical threshold curves for the terminal score s on the four synthetic equality-LP families. Each panel reports the fraction of instances satisfying $s \leq \tau$ as a function of the threshold τ .

Table 4 reports solver-reported runtimes; for time-limited runs, the time limit is used. This strict subset is small and should be read as a benchmark comparison rather than a comprehensive LP benchmark.

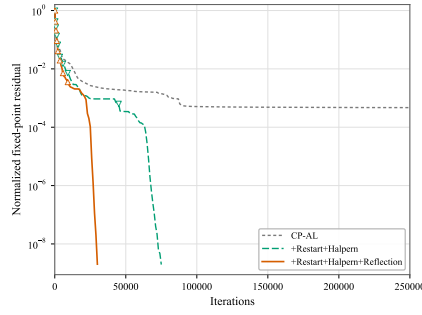
Finally, Figure 2 studies **tbfp-network** through an intra-method ablation. Each panel compares the base method, the restarted Halpern variant, and the restarted reflected Halpern variant for the same underlying map. The base

Table 4: Strict equality-form MIPLIB benchmark instances. Runs are performed without presolve, and entries are solver-reported runtimes in seconds. $\dagger$ marks a time-limited run.

Instance	cuPDLPx	HPR-LP	RHR-CP-AL
datt256	0.141	38.4	0.060
ds	4.929	858	9.524
ds-big	3600 $\dagger$	3600 $\dagger$	1903
eila101-2	2.779	452	4.638
eila76-2	2.307	300	2.753
ivu06	451	3600 $\dagger$	220
ivu06-big	2585	3600 $\dagger$	1286
ivu59	705	3600 $\dagger$	913
ns1952667	0.056	11.8	0.033
nw04	0.555	60.4	3.858
supportcase6	4.010	778	11.0
t1717	3.452	796	4.158
t1722	0.551	2.694	0.752
tbfp-network	0.432	59.8	0.885
Solved	13/14	10/14	14/14
Total time (s)	7359.1	17756.0	4359.5
SGM10 (s)	35.47	359.37	33.72



(a) PDHG.



(b) CP-AL.

Fig. 2: LP mechanism ablation on **tbfp-network**. Each panel compares the base method with **+Restart+Halpern** and **+Restart+Halpern+Reflection**. The vertical axis is each method’s fixed-point residual normalized by its initial value.

curves exhibit long stagnation phases, while the restarted variants enter a second phase in which the normalized fixed-point residual drops rapidly. This is the numerical counterpart of the restart/sharpness mechanism proved in the paper.

6.3 Quadratic Programming Experiments

We next consider convex QPs of the form

$$\min_{x \in \mathbb{R}^n} \frac{1}{2} x^\top H x + c^\top x \quad \text{s.t.} \quad Ax = b, \quad l \leq x \leq u, \quad (40)$$

with $H \succeq 0$. The QP experiments are deliberately scoped by algorithmic role rather than by exhaustively benchmarking every map in the framework. We separate three tests. The first is a geometry-controlled augmentation test: it identifies regimes where a positive augmentation parameter in **RHR-FA-CP** improves over the unaugmented PDHCG-type baseline. The second is a standard equality-only benchmark comparison against external QP solvers. The third is a linearization-cost test: it compares subproblem-based methods, including PDHCG-II, **RHR-CP-AL**, and **RHR-FA-CP**, with the explicit reflected linearized methods **RHR-Lin-PDHG** and **RHR-Lin-CP-AL**. We do not report a linearized **FA-CP** variant. Its linearization still leaves a nontrivial primal subproblem, whereas the point of the linearized comparison is precisely to test variants whose main update becomes explicit. Thus the synthetic and linearized experiments provide mechanism evidence, while the Maros–Mészáros subset serves as a standard-solver benchmark comparison.

For the subproblem-based QP variants, diagonal Hessian subproblems are solved by componentwise projection formulas, while sparse or low-rank non-diagonal Hessian subproblems are handled by the projected-gradient/Barzilai–Borwein inner solver used in PDHCG-II [22]. In **RHR-FA-CP**, this inner problem includes the augmented quadratic term. The linearized variants instead use explicit projected updates and therefore avoid these inner subproblem solves.

Geometry-driven synthetic QPs

The main QP experiment is designed around a single geometric question: when H provides little curvature in $\text{range}(A^\top)$ and the equality matrix A is ill-conditioned, can the augmented term in **RHR-FA-CP** improve an unaugmented PDHCG-type baseline? This is the regime in which a positive augmentation parameter is expected to help, since the additional $A^\top A$ -curvature acts precisely in the weak row-space directions.

All synthetic QPs in this subsection are generated from the same planted-KKT model. We first sample A , $H \succeq 0$, a primal point $x^* \in [l, u]$, equality multiplier y^* , and bound multiplier s^* satisfying the sign conditions at the active bounds. We then set

$$b = Ax^*, \quad c = A^\top y^* - Hx^* - s^*.$$

Thus the generated instance has a controlled KKT point, and different random families correspond to different parameter regimes of the same construction. To isolate the augmentation mechanism, we report targeted families in regimes where the preceding geometry suggests that positive augmentation

Table 5: Targeted equality-box QP mechanism experiment for positive augmentation in RHR-FA-CP. Each row aggregates total runtime over the 100 solved instances in the family.

Curvature	Constraint family	Activity regime	Baseline time	Best $\sigma > 0$	Positive- σ time	Speedup
Balanced	Ill-scaled	Moderately active	337s	10^{-2}	308s	1.09×
Row-space weak	Mildly near-dependent	Moderately active	438s	1	337s	1.30×
Row-space weak	Ill-scaled	Interior-dominant	512s	3×10^{-1}	489s	1.05×

should be useful: weak row-space curvature, ill-conditioned or nearly dependent equality constraints, and nontrivial box activity. We also include one balanced-curvature ill-scaled family as a control case. For the row-space-weak profile, we decompose $\mathbb{R}^n$ into $\text{range}(A^\top)$ and $\text{null}(A)$, draw the eigenvalues of H from $[10^{-8}, 10^{-6}]$ on $\text{range}(A^\top)$, and draw them from $[10^{-1}, 1]$ on $\text{null}(A)$. The balanced profile uses $[10^{-3}, 10^{-1}]$ on both subspaces. The equality matrices are either ill-scaled with weak dependence or mildly near-dependent with moderate row/column perturbations. The interior-dominant and moderately active regimes plant about 1% and 8% of the variables at each bound, respectively. We generate 100 instances per reported family, with a 40/40/20 split over $(m, n, \text{density}) = (60, 160, 0.060)$, $(100, 240, 0.040)$, and $(160, 320, 0.025)$.

The comparison within this experiment is deliberately narrow. We keep the same restarted reflected Halpern wrapper and compare the unaugmented PDHCG-type baseline with positive- σ RHR-FA-CP variants. This directly tests the effect of augmentation while keeping the outer acceleration and residual criterion fixed. The candidate set is

$$\sigma \in \{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 3 \times 10^{-1}, 1\}.$$

The tolerance is 10^{-6} , and the time limit is 30 seconds per instance. For each reported family, the selection rule is fixed in advance: maximize the solved count first and break ties by total runtime. Table 5 reports the resulting mechanism comparison. In the table, “Baseline total time” is the total wall-clock time over the 100 instances for the unaugmented baseline, “Positive- σ total time” is the corresponding total time for the selected positive value, and the speedup is their ratio.

Figure 3 provides two mechanism views. To avoid mixing two different notions of improvement, the left panel focuses on the same targeted families as Table 5. The right panel shows a representative row-space-weak, ill-scaled, interior-dominant instance drawn from the same planted-KKT generator.

In this representative instance, the unaugmented PDHCG-type run reaches 10^{-6} after 12800 iterations, whereas $\sigma = 10^{-2}$ and $\sigma = 0.3$ reach the same tolerance after 5250 and 4100 iterations, respectively. This trace is not used as a broad benchmark claim; it is included to make visible the mechanism

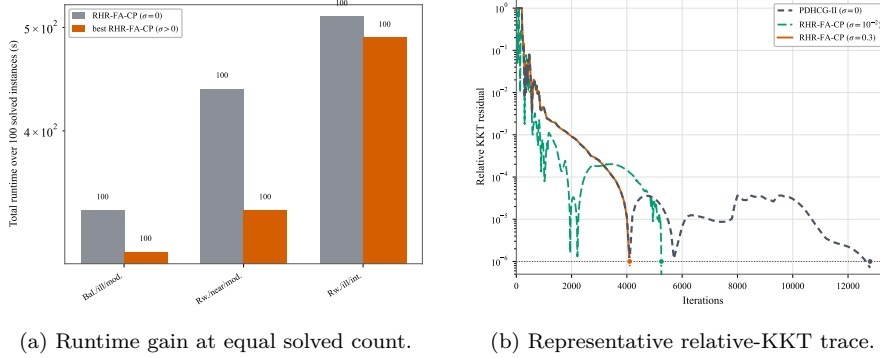


Fig. 3: Mechanism trace for positive augmentation in the random equality-box QP experiment. Panel (a) compares the unaugmented PDHCG-type baseline with the best positive- σ RHR-FA-CP variant on the targeted families; numbers above the bars are solved counts. Panel (b) shows a row-space-weak, ill-scaled, interior-dominant instance for zero augmentation, $\sigma = 10^{-2}$, and $\sigma = 0.3$.

Table 6: Equality-only Maros–Mészáros QP subset. The best solved count is bolded, and timing entries are bolded when they are best among methods that solve all 50 instances.

Method	Solved	Total time	Median time	SGM10
RHR-FA-CP	50/50	1033.00s	2.00s	7.19s
PDHCG-II	50/50	1057.00s	3.00s	7.41s
HPR-QP	50/50	876.23s	3.90s	8.46s
PDQP	48/50	3183.89s	0.91s	10.89s

suggested by the factorized experiment, namely that positive augmentation can shorten the terminal transition phase caused by weak or ill-conditioned equality-constraint geometry.

Equality-only Maros–Mészáros subset

We next report a standard benchmark comparison on the equality-only subset of the Maros–Mészáros QP repository [29]. The subset contains the 50 instances whose constraints are pure equalities with variable bounds; no slack lifting is needed to match the equality-box form in (40). We compare our selected RHR-FA-CP configuration with PDHCG-II [22], PDQP [28], and HPR-QP [10]. Presolve is disabled for all solvers, the tolerance is 10^{-6} , and the time limit is 1000 seconds per instance. The RHR-FA-CP configuration is selected by the same solved-count-then-runtime rule used above. Unsolved runs are charged with the time limit when computing total time and SGM10.

This benchmark serves as a standard-solver comparison rather than as the main test of the augmentation mechanism. It shows that the selected

Table 7: Unified planted-KKT large-scale QP regime for testing explicit linearized variants (1000 instances). All methods solve all instances; SGM10 is the shifted geometric mean time in seconds, and the last column reports the median cost per 10^4 iterations.

Method	Solved	Total runtime	SGM10	Median sec. per 10^4 iters.
PDHCG-II	1000/1000	3485s	3.37	43.80
RHR-CP-AL	1000/1000	3546s	3.43	37.04
RHR-FA-CP	1000/1000	2885s	2.79	32.79
RHR-Lin-PDHG	1000/1000	2526s	2.42	8.70
RHR-Lin-CP-AL	1000/1000	3465s	3.32	9.76

RHR-FA-CP method is competitive with mature QP solvers when presolve is disabled; the synthetic experiment above isolates the geometric mechanism behind augmentation.

Linearized variants

The third QP regime uses the same planted-KKT generator to test the numerical meaning of the linearized theory. Among the linearized methods we include only RHR-Lin-PDHG and RHR-Lin-CP-AL, because these are the variants for which linearization turns the main step into an explicit update. The subproblem-based PDHCG-II, RHR-CP-AL, and RHR-FA-CP rows are included as subproblem-based baselines. A linearized FA-CP variant is not reported: even after linearization it still requires solving a nontrivial primal subproblem and therefore does not isolate the explicit-update mechanism. The aggregate comparison is reported in Table 7.

The linearized test regime keeps A well scaled and weakly dependent, uses sparse moderate-curvature Hessians, and increases the problem sizes to make the cost of inner subproblem solves visible. We generate 1000 instances in four families: banded or sparse Hessian structure, crossed with interior-dominant or moderately active bounds. The size split is 40/40/20 over

$$(m, n, \text{density}) \in \{(300, 1200, 0.010), (520, 2600, 0.0045), (650, 3600, 0.0032)\}.$$

All methods use tolerance 10^{-5} and a 60 second per-instance time limit. The augmented parameter for RHR-FA-CP is fixed at $\sigma = 10^{-3}$. For RHR-Lin-CP-AL, we use $\sigma_0 = 10^{-1}$. These values are chosen once before the benchmark and are not tuned instance by instance.

The point of this experiment is not that explicit linearization dominates every QP geometry. Rather, it identifies the computational regime targeted by the linearized theory: when the Hessian is cheap to apply and the primal subproblem solve is no longer negligible, the explicit update can have much lower per-iteration cost. In this regime, RHR-Lin-PDHG gives the smallest total time, and both RHR-linearized methods reduce the median cost per 10^4 iterations by roughly a factor of four relative to the subproblem-based methods.

7 Conclusion

We developed a unified augmented primal–dual framework for linearly constrained composite convex problems. The analysis follows a chain from augmented primal–dual maps to exact dPPM and PFBS representations, reflected Halpern KKT rates, identification of shadow sequences, and sharpness-based restart. The exact implicit scheme admits an exact degenerate preconditioned proximal-point reformulation, whereas the linearized scheme admits an exact preconditioned forward–backward splitting reformulation. These two operator representations support a common reflected Halpern analysis: they yield convergence to KKT points and nonergodic $O(1/k)$ bounds for KKT residuals and objective gaps. A one-dimensional quadratic example shows that this global residual rate is worst-case tight.

The main local structural point is that the correct finite-identification object for reflected Halpern trajectories is the shadow sequence, not the anchored Halpern state. Under an additional exact affine-face hypothesis, the identified shadow points admit an exact reduced residual identity, a reduced Jacobian criterion for local sharpness, and a transfer from reduced error bounds to the original augmented KKT mapping. Building on these ingredients, we proved fixed-frequency and adaptive restart results under sharpness on the relevant trajectory set. When this sharpness is supplied by local error bounds, the resulting linear convergence is a local or tail statement in the positive definite metric setting.

The numerical section follows this structure: equality-constrained box LPs and equality-constrained box QPs are treated separately, and each comparison uses members of the unified family that are computationally meaningful for the corresponding representation. This organization helps separate structure-induced performance differences from effects caused purely by problem representation or by inner-solver cost.

Several directions remain open. On the theoretical side, it would be desirable to weaken the exact affine-face hypothesis, replacing the exact residual identity by polyhedral error-bound or metric-subregularity arguments for ℓ_1 and more general polyhedral structures, and to extend the restart verification results beyond the positive definite metric setting. On the computational side, natural next steps include improving the GPU realization of the lifted augmented variants and broadening the benchmark campaigns beyond the equality-form LP/QP regimes considered here.

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